MIL-HDBK-226(NAVY)
17 JUNE 1968

MILITARY STANDARDIZATION HANDBOOK

DESIGN DISCLOSURE FOR SYSTEMS AND EQUIPMENT

MISC

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NAVAL SHIP ENGINEERING CENTER

WASHINGTON, D. C. 20360

MIL-HDBK-226(NAVY)
DESIGN DISCLOSURE
FOR SYSTEMS AND EQUIPMENT

1. This standardization handbook was developed by the Naval Ship Engineering Center with the assistance of the Naval Applied Science Laboratory in accordance with established procedure.

2. This publication was approved on 17 June 1968 for printing and inclusion in the military standardization handbook series.

3. This standardization handbook sets forth the requirements for preparing design disclosures for equipment and systems.

4. It is the intent to review this handbook periodically to insure its completeness and currency. Users of this document are encouraged to report any errors discovered and any recommendations for changes or inclusions to the Naval Ship Engineering Center, Department of the Navy, Washington, D. C. 20360.

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FOREWORD

In order to enhance the evolution of effective military systems, this handbook has been issued to provide a clear understanding of the development of these systems at all functional levels and at all life cycle phases.

The design disclosures required by this handbook will comprise a uniform set of lucid, comprehensive technical communication techniques which enable technical control by management, effectiveness modeling and analysis, and design assurance for performance, reliability, maintainability and human factors.

Hence, the design disclosure set becomes the fundamental vehicle for transmitting design information between the many activities that must jointly work toward one objective; namely, the deployment and utilization of an effective system. This technical communication media comprehensively maps the system equipment design in a concise and understandable manner.

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1. SCOPE

1.1 This handbook sets forth a methodology for preparing equipment and systems for design disclosures. These disclosures will show the general and specific formulation of the development and design of the system or equipment. The disclosure will be the basic contractor system description to enable military program managers and procurement agencies to effectively analyze, review, and monitor the performance, maintainability, and reliability design of systems or equipments throughout the development life cycle. This analysis will be accomplished by calling out, in addition to this basic document, (a) detailed requirements and (b) data elements via DD1423. The contractor shall be required to key all data elements to this system disclosure through appropriate identifier coding.

2. REFERENCED DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this handbook to the extent specified herein.

GOVERNMENTAL

SPECIFICATIONS
MILITARY

MIL-D-1000	- Drawings, Engineering and Associated Lists
MIL-D-5480	- Data, Engineering and Technical: Reproduction Requirements for.
MIL-Q-9858	- Quality Program Requirements
MIL-M-24100	- Manuals, Orders and Other Technical Instructions, for Equipment and Systems
MIL-H-46855	- Human Engineering Requirements for Military Systems, Equipment and Facilities

STANDARDS
MILITARY

MIL-STD-12	- Abbreviations for Use on Drawings and in Technical-Type Publications
MIL-STD-15-2	- Electrical Wiring Equipment Symbols for Ships' Plans Part-2
MIL-STD-100	- Engineering Drawing Practices
MIL-STD-280	- Definition of Terms for Equipment Divisions
MIL-STD-470	- Maintainability Program Requirements (For Systems and Equipments)
MIL-STD-471	- Maintainability Demonstration
MIL-STD-721	- Definitions of Effectiveness Terms for Reliability, Maintainability, Human Factors, and Safety

HANDBOOKS
MILITARY

MIL-HDBK-217	- Reliability Stress and Failure Rate Data for Electronic Equipment
MIL-HDBK-472	- Handbook of Maintainability Prediction

PUBLICATIONS
MILITARY

NAVSHIPS 91727	- Electronic Test Equipment Application Guide
NAVSHIPS 93820	- Handbook for the Prediction of Shipboard and Shore Electronic Equipment Reliability
NAVSHIPS 94324	- Maintainability Design Criteria Handbook for Designers of Shipboard Electronic Equipment
DD Form 254	- Securities Requirement Check List
DOD 5220.22-M	- Industrial Security Manual for Safeguarding Classified Information

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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NON-GOVERNMENTAL

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd, Chicago, Illinois, 60606)

UNITED STATES OF AMERICA STANDARDS INSTITUTE (USAS)

Y14.15, Section 15	- Electrical Diagrams
Y32.2	- Graphic Symbols for Electrical and Electronic Diagrams
Y32.14	- Graphic Symbols for Logic Diagrams
Y32.16	- Electrical and Electronic Reference Designations

(Applications for copies should be addressed to United States of America Standards Institute, 10 East 40th Street, New York, New York 10016)

3. DEFINITIONS

3.1 The following definitions apply to hardware and functional identities within the design disclosure package. The hardware definitions are in accordance with MIL-STD-280.

(a) Hardware
(1) Part. One piece, or two or more pieces joined together which are not normally subject to disassembly without destruction of designed use. (Examples: electron tube, resistor, mica capacitor, audio transformer, microcircuit.)
(2) Subassembly. Two or more parts which form a portion of assembly or a unit replaceable as a whole, but having a part or parts which are individually replaceable. (Examples: I-F strip, terminal board with mounted parts.)
(3) Assembly. A number of parts or subassemblies or any combination thereof joined together to perform a specific function. (Example: audio-frequency amplifier.)
(4) Unit. An assembly or any combination of parts, subassemblies, and assemblies mounted together, normally capable of independent operation in a variety of situations. (Examples: electronic power supply, radio receiver.) Note: The size of an item is a consideration in some cases.
(5) Group. A collection of units, assemblies, or subassemblies which is a subdivision of a set or system, but which is not capable of performing a complete operational function. (Examples: antenna group, indicator group.)
(6) System-equipment(set). A unit or units and necessary assemblies, subassemblies, and parts connected or associated together to perform an operational function. (Examples: radar set, sonar set, etc.)
(b) Functional
(1) Functional element. One or more parts which perform(s) a functional operation. Also, the fewest number of parts that can perform a functional operation. (Examples: amplifier, emitter follower, differentiator, current limiting resistor, etc.)
(2) Composite. Two or more functional elements which together perform a functional operation. Also, any level of functionalization between functional elements and functions. Functions are defined below. (Examples: power supply, regulator, timer, counter, etc.)

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(3) Function. Any combination of two or more functional elements or composites. Also, functions are natural operational breakdowns of a system-equipment. (Examples: transmitter, receiver, video processor, indicator, etc.)

(4) Functional entity. Functional element, composite, function or any combination thereof.

(c) Other

(1) Terminal event. A terminal event is a system-equipment output. In addition, it is an event which must be identified by both a specific measurement point and a specific mode of operation. Example: PPI output (track mode.)

(2) Design. The process of selecting the means and contriving the elements, steps and procedures for producing what will adequately satisfy some need.

(3) Disclosure. A statement or description of an invention or device and its method of operation.

4. GENERAL REQUIREMENTS

4.1 General. The requirements for design disclosures will vary with the size and type of equipment and with each phase of the equipment life cycle. Since the development life cycle also varies from equipment to equipment, it is not practical for this handbook to fix the requirements for all procurements. Rather, it will be the responsibility of the procuring agency to tailor the requirements of this handbook to the specific needs of a particular equipment by specifying in the procurement contract modifications to the requirements listed in the following paragraphs. Detailed data requirements shall be specified in DD1423 and keyed to this disclosure requirement.

4.1.1 Instructions and examples provided in this handbook establish techniques for the preparation of design disclosures. Contractors required to submit design information according to this handbook shall employ the techniques described and exemplified herein, as appropriate for each phase of the system life cycle. At all times, the objective shall be to provide clear, complete, and concise documentation of the results of the design process. Where new or unique situations are encountered which are not specifically covered by this guide the principles of formating shall be extrapolated by the contractor. In all cases, the procuring agency retains the right of inspection and approval of the final product.

4.1.2 Design formating discloses the mechanization in a functional/physical oriented scheme. Although the design disclosures do not form an analytic presentation, they do form a disciplined foundation of information upon which objective and formalized analysis can be built. The design disclosures consolidate, in a unified scheme, basic data required in a variety of engineering disciplines. They form a single source document containing performance, reliability, maintainability and human factors information. Thus, the design disclosure set is a common technical language for prime equipment contractors, support equipment contractors, Navy procurement activities, and system analysis personnel.

4.2 Generation and use. Design disclosures are generated throughout the system development life cycle. As the design matures, the design disclosures similarly grow in content and detail. At specific points of the design process, information is submitted according to the specified format. This information is used to review the design of the subject equipment, and to coordinate that design with the designs of other equipments to which it must mate. This technical control provides a mechanism for assuring the performance effectiveness of the subject equipment and a guarantee of compatibility with interfacing and associated equipments.

4.2.1 Design disclosures are created for the benefit of the military as well as the contractor. It is expected to be of value to the contractor because the increased design visibility provided by the design disclosure set forces the engineering and scientific efforts to proceed with direction and purpose. New contractor personnel quickly comprehend the system, and the many design, support, and staff functions may be performed using a common source document.

4.2.2 Many of the engineering documents currently generated (functional block diagrams, schematics, descriptive materials, etc.) may be replaced by the corresponding types of design disclosures but other drawings, arrangement drawings, interface control drawings, wiring diagrams, mechanical drawings and others as specified in MIL-D-1000 which the disclosures do not replace, can be referenced to the disclosures. Near the final stages of development, the design disclosures are suitable for direct use in technical training and maintenance support documentation efforts.

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4.2.3 The design disclosure set will lead to quicker recognition of required design changes or improvements. As it is an evaluation tool for the Navy, it is a design tool for the contractor. Thus, contractors are encouraged to incorporate the disclosure generation as an integral function of the engineering activity. If the disclosure preparation is separated organizationally, a less efficient process may result.

4.2.4 While the design disclosure handbook has been issued for implementation on new system development programs, it also may be used effectively on secondary procurements for equipments already deployed, but which need redesign to keep up with the state of the art. Such efforts frequently involve redesign studies leading to a comprehensive evaluation and design review before entering production. Here, too, the design disclosure techniques may be employed as a technical reporting scheme.

4.2.5 This handbook has been prepared recognizing that all procurements have differing requirements. It has thus been identified as a general handbook. Flexibilities have been built in to permit adaptation of the techniques to a wide variety of applications. It is expected that the specific requirements for each procurement will be generated as a normal consequence of the contracting process.

5. DETAILED REQUIREMENTS

5.1 Introduction. The design disclosure concept uses four basic techniques to present design information. These techniques are identified as follows:

(a) Blocked schematics
(b) Detailed block diagrams
(c) Blocked text
(d) Design outlines

To the extent determined by the contracting agency these techniques are applied during each phase of a system's or equipment's development to disclose its design characteristics. At significant milestones, the contractor will supply a design data package to the military. Design disclosures form part of this package and are procured by the means of this handbook. Design disclosures may be used individually for in process type reviews or assembled into a unified design data package for major design reviews. The assembled product completely and concisely defines the design at all levels. The design disclosure package, by itself or as part of a total design package, becomes a highly effective tool to be used by many interfacing technical groups.

5.2 Content. The intent of this paragraph is to specify for contractual purposes the types of design disclosures required. Each type of disclosure is described in detail in the appendix.

5.2.1 Master block diagram text. One master block diagram text shall be provided to:

(a) Define the major functions.
(b) Disclose the information flow between these several functions.
(c) Provide general theory of operation descriptions for the functions.
(d) Define the areas of hardware containment for the various functional entities of the system-equipment.

5.2.2 Master design outline. One master design outline shall be provided for each system equipment to:

(a) Summarize the logical dependency structure of the entire system-equipment.
(b) Disclose the relationships between functional entities.
(c) Provide detailed event and signal identifications for all system-equipment inputs and outputs.
(d) Consolidate a wide spectrum of performance, reliability, and maintainability data on a single format.

5.2.3 Intermediate block diagram text. Intermediate block diagram texts shall be provided to insure smooth and comprehensive transition between detailed block diagram and the master block diagram when a system-equipment is sufficiently large and complex to warrant these diagrams. The intermediate block diagram/text provides intermediate level text linking relating detailed block diagrams to the master block diagram/text. They consolidate detailed performance, reliability, and maintainability data associated with the detailed block diagrams into higher level data.

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5.2.4 Intermediate design outline. Intermediate design outlines shall be provided to insure smooth and comprehensive transition between the many detailed design outline parts and the master design outline when a system-equipment is sufficiently large and complex to warrant these diagrams. The intermediate design outlines provide intermediate level dependency structure coverage linking the detailed design outline parts to the master design outline. Detailed performance, reliability, and maintainability data associated with the detailed design outline parts are consolidated into higher level data. See section 2 for applicable reliability and maintainability standards.

5.2.5 Detailed block diagrams. Detailed block diagrams depict the mechanization scheme of the system equipment. Detailed block diagram coverage shall be provided for each function throughout a system-equipment. For simple system-equipments, a detailed block diagram may cover more than one function. A function is a natural functional breakdown of the system-equipment such as the power distribution, transmitting, receiving, and timing functions. The basic building blocks of the detailed block diagram are the functional element and composite levels of functional entities (e.g. emitter follower, flip-flop, voltage regulator, etc.)

5.2.6 Detailed block diagram tests. One blocked text diagram shall be provided for each detailed blocked diagram.

5.2.7 Power distribution diagrams. The number of power distribution diagrams depends on system-equipment complexity and design. Generally, there shall be a power distribution diagram for each major hardware item and one diagram showing distribution between cabinets. However, the basic objective is to limit the number of diagrams to as few as possible.

5.2.8 Power distribution diagram texts. One blocked text diagram shall be provided for each power distribution diagram.

5.2.9 Detailed design outline. One detailed design outline shall be provided for each system-equipment. This design outline is divided into parts, each part corresponding to a single function (a function is defined in section 3.)

5.2.10 Blocked schematics. One blocked schematic shall be provided for each unit, assembly, etc. Blocked schematics will not be required for cabinets, racks or enclosures which function primarily as a mounting and interconnecting device for removable assemblies. Any parts or connectors associated with a device of this nature will be shown in schematic form on the appropriate detailed block diagram or power distribution diagram. On small system-equipments, a single blocked schematic may be prepared, if possible, which will contain both assembly and interconnection information.

5.2.11 Blocked schematic texts. One blocked text diagram shall be provided for each blocked schematic.

5.2.12 Operational sequence diagrams. If required by the contracting agency, operational sequence diagrams shall be provided in accordance with MIL-H-46855.

5.2.13 Front matter. Front matter shall include a title page, table of contents, and a description of all special symbols and codes used on the disclosures.

5.2.14 Supplemental information. The number of supplemental sheets is determined by the amount of support information required to backup the disclosures. Support information includes such items as: (a) system-equipment description (b) reliability block diagrams, (c) model calculations, (d) terminal event calculations, (e) parts location data, (f) alignment procedures, (g) maintenance procedures, (h) data sources, etc. as required separately by DD1423, (i) packaging data.

5.2.15 Life cycle implications. The cognizant procuring agency will specify the schedule and quantity of deliveries to be made at various points in the life cycle.

5.3 Format.

5.3.1 Diagram arrangement. Diagrams shall be arranged in the order of the most general data to the most detailed data. Blocked text shall always be on the page facing the diagram being described. For use in operating and technical manuals, arrangement and details shall be as specified in applicable specifications.

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5.3.2 Abbreviations. Abbreviations for use on illustrations shall conform to MIL-STD-12.

5.3.3 Graphic symbols. Use graphic symbols for electronic diagrams from USAS Y32.2, electrical wiring equipment symbols from MIL-STD-15-2, and logic symbols from USAS Y32.14. For required details on the application of graphic symbols, refer to USAS Y14.15.

5.3.4 Reference designations. Electronics reference designations shall conform to USAS Y32.16.

5.3.5 Functional coding. A complete functional coding scheme shall be employed to uniquely define all functional entities in the system-equipment hierarchy structure. This scheme shall be compatible with requirements for automatic data processing.

5.3.6 Security classification. The security classification of a design disclosure set shall be as designated by the cognizant government activity. Security requirements check list DD Form 254, which constitutes a part of the contract for all classified material, identifies and indicates the classified features of the system-equipment. All pages of the design disclosure set shall be marked in accordance with the Industrial Security Manual for Safeguarding Classified Information, DOD 5220.22-M.

5.4 Drafting standards.

5.4.1 Layout. Recommended layout shall be a function of the equipment to be documented. A page size approximately 15 by 20 inches (reduced) or smaller is desirable. However, a maximum page size (reduced) of 15 by 35 inches is acceptable. Layout should be such that the percentage of reduction is minimized. Reduction shall never exceed 60 percent. For early phase deliveries, upsize (original) art may be used.

5.4.2 Lettering. Lettering typography shall be as set forth in table I. It is recommended that lettering be typed on transparent material (Par-Tone Products Printable Friskits (copy-zip), or equivalent) with adhesive back to assure a permanent attachment to the original art. For early phase deliveries, legible pencil art may be used.

Table I - Typography and lettering

Use	Face and point size	Case
Security classification	Same type family as body text 14 point final size. (Final size)	Upper
Chapter and section titles	12 point extra bold Sans Serif (Final size)	Upper
Note (word only)	12 point extra bold Sans Serif (Final size)	Upper
Caution (word only)	12 point extra bold Sans Serif (Final size)	Upper
Warning (word only)	12 point extra bold Sans Serif (Final size)	Upper
Figure titles	12 point extra bold Sans Serif (Final size)	Upper
Body and all other written matter and lettering	Eton bold, varityper 1040-12A or equivalent reduced never more than 60 percent and desirably not more than 50 percent in final size.	Upper

Note: When a page is to contain large areas of written matter on a white background (no shading) and the percentage of reduction will not be greater than 30 percent in final size, 10 point modern type face, both upper and lower case characters may be used.

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5.4.3 Line artwork. Line artwork shall be prepared with line weights of sufficient strength to reproduce sharply and clearly at the final reproduction size. Line artwork shall be prepared such that ozalid or equivalent copies of the original artwork can be obtained.

5.4.4 Legibility. Clarity shall be in accordance with requirements of MIL-D-5480.

5.5 Reproduction. When formalized production is required, design disclosures may be reproduced either by the offset process or by the diazo (blackline) process or equivalent. The process to be used shall be specified in the procurement contract.

5.5.1 Diazo process. Reduced size copies (15 by 35 inches or smaller) of the design disclosure set shall be provided for the reviewing agency, reproduced by the diazo process or equivalent. Background and contract shall conform to commercially accepted standards with all lettering lines and symbols clearly legible.

5.5.2 Offset process. Reduced size copies (15 by 35 inches or smaller) of the design disclosure set shall be provided for the reviewing agency, reproduced by the offset process. Letters, lines and symbols shall be of uniform contrast throughout the final printed material. Blurred or smudged printing or drop-out characters, lines or letters shall be cause for rejection.

5.5.2.1 Page stock. Page stock shall be E-40 chemical wood map paper lithographic finish, or equivalent, substance weight not less than 44 pounds per 1000 sheets, basis 17 by 22 inches.

5.5.2.2 Shade option. At certain major milestones in the equipment life cycle, the procuring agency may specify in the procurement contract the use of shading to define functional and hardware boundaries. Section 40.1 of the appendix defines the requirements of the technique and describes in detail all of the necessary processes.

5.5.3 Covers. Covers shall be provided for design disclosures reproduced by diazo or offset process or equivalent. Weight, and composition of the material shall be left to the contractor's discretion unless otherwise specified in the procurement contract. Covers for unclassified documents may be of any color except green, yellow or pink. Covers for CONFIDENTIAL documents shall be green, covers for SECRET documents shall be yellow, and covers for TOP SECRET documents shall be pink.

5.5.4 Binding. The design disclosure set shall be prepared in looseleaf form to facilitate the insertion of replacement pages. Commercial individually hinged metal ring (or plastic equivalent) fasteners are to be used. The disclosure set may be either top or side edge bound.

5.6 Quantity. The quantity of design disclosure sets to be delivered to the reviewing agency will be as set forth in the contract.

5.7 Reproduction rights. Reproduction rights shall be as established by the provisions of the contract pertaining to data or copyright.

5.8 Quality assurance provisions.

5.8.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

5.8.2 Quality assurance program requirements. The supplier shall provide and maintain a written quality assurance program acceptable to the Government. The program shall be delineated in sufficient depth to assure that adequate and accurate data is being presented, and that the data is in accordance with the requirements of this handbook. MIL-Q-9858 may be invoked all or in part by the contracting agency.

5.8.2.1 Acceptance of the quality assurance program by the Government is not intended to preclude additions, refinements, or changes to the written quality assurance program where evidence indicates the need for such action by the Government activity.

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5.8.3 In-process reviews. Design disclosures shall be subject to in-process review prior to publishing. Reviews may be conducted at the contractor's facility or his source facility at any time during the development of the disclosures. However, reviews will be held prior to the preparation of final reproducible copy.

5.8.3.1 In-process reviews will be executed by the Government (Government's participation at its option) and the contractor in a timely manner. The contractor shall be responsible for the establishment of the schedule that will permit the government to review the contractor's mode of preparation, compliance with specifications, technical content, problem areas, etc.

5.8.3.2 In-process reviews will normally be initiated at the discretion of the cognizant service; however, reviews may be requested by the contractor when assistance or clarification is desired. Government intentions related to reviews will be submitted to the contractor prior to visitation.

5.8.3.3 The cognizant Government activity shall designate the organizations and personnel authorized to participate in the review.

5.8.4 Validation. Validation performed under this handbook shall constitute an integral part of the quality assurance program requirements of the basic contract. The object of validation is to assure that the contractor has conformed to this handbook.

6. PREPARATION FOR DELIVERY

6.1 Packaging and packing. Containers shall comply with the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation.

7. NOTES

7.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this handbook.
(b) Security classification (see 5.3.6).
(c) Types of disclosures and their content (see section 5).
(d) Reproduction process (see 5.5).
(e) Shade option (see 5.5.2.2).
(f) Quantity (see 5.6).

7.2 Handbook interpretation. Any area of the handbook which a supplier feels is vague, should be brought to the attention of the cognizant activity prior to completion of the contract.

Review activities:
Navy - SH, YD, AS, OS, EC
User activity:
Navy - MC

Preparing activity:
Navy - SH
(Project-Misc-N515)

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APPENDIX

DISCLOSURE GUIDE

10. DELIVERABLE ITEMS. The material included in this appendix is a mandatory part of this handbook.

In order to provide a clear understanding of the components of the design disclosure package, the basic disclosures are presented in 10.1 through 10.5 without specific reference to life cycle application. Paragraph 10.6 will describe how the basic disclosures are applied to specific life cycle phases and to specific levels within a system hierarchy.

10.1 Blocked schematics (see figure 1). Blocked schematics represent the most detailed level of design disclosure. The difference between a blocked schematic and a conventional schematic is primarily one of circuit layout and circuit definition. On blocked schematics, emphasis is placed on grouping circuit parts by circuit function in order to provide precise stage definition to the part level. For example: All of the parts which work together to form a blocking oscillator are enclosed by a light solid line. The benefit which is derived from proper grouping and layout is immediately obvious. In most cases, the user merely has to glance at the circuit in order to understand its operation. In cases of new or unusual circuits, proper layout is the first step toward the understanding process. In addition, an identifier code is assigned to each functional grouping to uniquely identify it in the system structure. Additional features of blocked schematics are coded signal lines and the separation of power entities from electronic functional entities. Separation of the power entities from the electronic functional entities helps a great deal in providing left to right functional signal flow and the simplification of the functional blocks.

Schematics are normally prepared with one blocked schematic for each proposed hardware assembly. If packaging details are not known, blocked schematics are generated by function only and if necessary, are subfunctionalized in some convenient manner to allow presentation on standard size format sheets. Text coverage of a blocked schematic is provided in the form of a blocked text diagram (see 10.3). The blocked text diagram is always placed on the facing page to the schematic and takes the form of the schematic, however, now the area that is taken up by the circuit symbols is replaced with the descriptive text. Figure 1 is an example of a blocked schematic. See 20.10 for a detailed description of the blocked schematic format.

10.2 Detailed block diagrams (see figure 2). Detailed block diagrams show in symbolic form interrelationships of the circuits which are required to perform a function. The detailed block diagram consists of symbols representing functional entities defined on the block schematics and are directly referenced to the blocked schematics by means of the identifier codes. Each symbol on the detailed block diagram is identified by a circuit identifier code which ties the particular symbol to the entire system structure. The code tells the reader that the symbol represents a particular functional entity. The detailed design information on this functional entity can be found by locating it on the appropriate blocked schematic. On the blocked schematic the functional entity is identified by the same code, and all of the parts which make up the entity are enclosed by a light solid line. Thus, it can be seen that there is a precise definition between the detailed block diagram and the blocked schematic. Precise definition also exists between the detailed block diagram functional entities and the upper level disclosures.

In addition to precise functional definition, detailed block diagrams define the proposed hardware boundaries, show complete interconnection information and use special symbols to code signal lines. Layout is of prime importance and rigid rules are specified to insure the best possible signal flow.

In general, a detailed block diagram is generated for each major functional division (function) as defined on an upper level blocked diagram/test disclosure. This will be discussed in more detail in section 20. Detailed block diagrams pictorially describe functional operation, define the proposed hardware boundaries and serve as the basis for the preparation of detailed design outlines.

Text coverage of a detailed block diagram is provided in the form of a blocked text diagram (see 10.3). The blocked text diagram is always placed on the facing page to the detailed block diagram and takes the form of the diagram, however, the area taken up by the symbols is replaced with the descriptive text.

Figure 2 is one example of a detailed block diagram. See 20.5 for a detailed description of the detailed block diagram disclosure.

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10.3 Blocked text (see figures 3 and 4). Blocked text is a combination block diagram and text on one disclosure. Blocks representing functional entities are connected in block diagram fashion by signal lines. Text within the blocks describes the operation of the functional entities and can accommodate additional information such as reliability, maintainability, logistics, human factors and other as required.

At the more general levels of disclosure, each of the blocks represents a major functional division of the system or equipment. Thus, the diagram provides a precise functional definition. This diagram may also be used to describe the overall characteristics including performance data and functional interrelationships.

At the more detailed levels, blocked text diagrams serve as companion diagrams to detailed blocked diagrams and blocked schematics. Circuits appear on the detailed block diagrams and blocked schematics in symbolic form while on the companion blocked text diagrams the symbol groupings are replaced by descriptive text. The blocked text is always placed on a facing page to its companion detailed block diagram or blocked schematic so that the symbolic representation of a design and its description are always together.

Figures 3 and 4 are examples of the three types of blocked text disclosures. See 20.1, 20.6 and 20.11 for a detailed description of each of these disclosures.

10.4 Design outlines (see figure 5). The operation of a system-equipment can be described in terms of the inputs supplied, the functions used, and the output events that result. When functional entities are interconnected to produce complex functions, the output of one becomes the input to the next, and so on. As more and more functional entities are interconnected, the interrelationships become involved, and are difficult to define with conventional diagramming techniques. The design outline is a logical model which portrays complex dependency chains in a clear and concise manner. By the use of a few special symbols, the interrelationships are developed witch visually symbolize the dependency structure of the system-equipment.

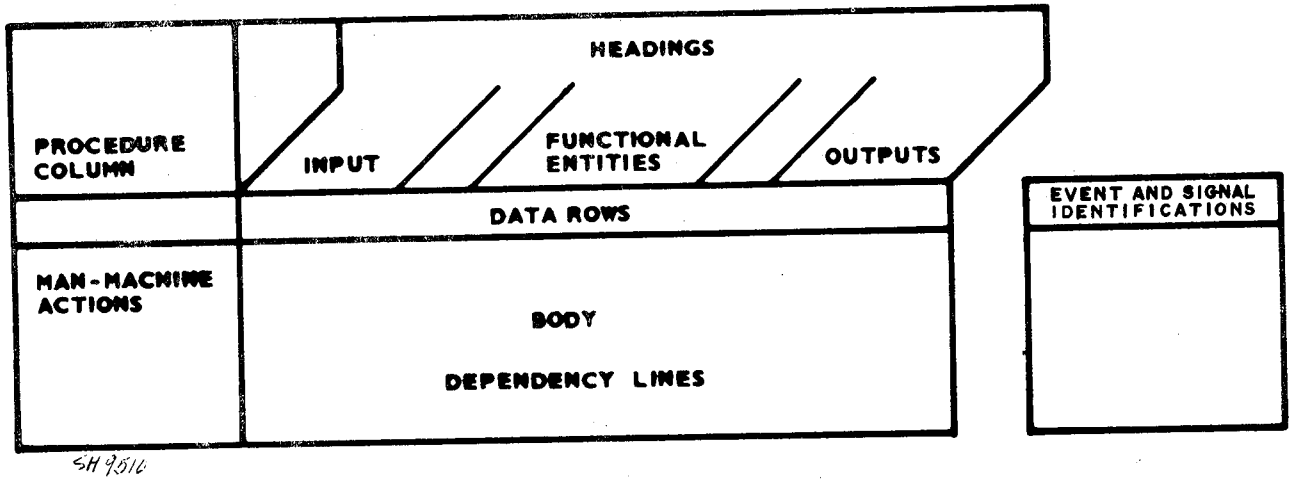
Because the design outline is a logical model of the functional design, it serves as a valuable tool for analytic techniques which can be performed to determine design adequacy.

Design outlines can be prepared to show detailed design disclosure or to summarize the overall design of the system-equipment. Generally, a master design outline is prepared to show the overall dependency structure of the functions of the system-equipment, and detailed design outlines to show the dependency structure of the lower level functional entities. It is not expected that design outlines be prepared to indicate circuit part relationships.

Figure 5 is an example of a master design outline. See 20.2, 20.4, 20.9 and 30.7 for a detailed description of design outlines.

10.4.1 Structure. All design outlines are constructed in a chart form consisting of four areas: (a) body (dependency structure), (b) headings, (c) operation column, and (d) event identification and data. The BODY of a design outline symbolically depicts the interrelationships among functional entities and their associated input and output events. The HEADINGS (located at the top of the chart) precisely identify each functional entity and its input and output events. The OPERATION COLUMN (located at the left of the chart) identifies all man-machine actions (turn-on, check-out, operations, etc.) required to effect utilization of the system or equipment. The event identifications (normally located to the right of the chart) precisely describe the events depicted in the chart body. Data rows (located directly below the headings) provide basic reliability and maintainability information associated with the functional entities and a reference to performance data associated with the events. These rows may also include cost, spares data, and other pertinent information.

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10.4.1.1 Body. The body of a design outline consists of a series of horizontal dependency lines. Each dependency line consists of basic symbols used to relate a functional entity or group of functional entities to their associated inputs and outputs. The organization of the dependency lines in the chart body portrays the energy flow through the functional entities depicted on the chart.

10.4.1.1.1 Symbols. The dependency line is constructed using three basic symbols. They are:

- □ to indicate an action or availability of one or more events resulting from the proper operation of the functional entities associated with the event.
- ● to indicate a functional entity or a group of functional entities.
- ▲ to indicate dependency upon another event.

Variations of these basic symbols will be discussed in 30.7.

10.4.1.2 Headings. The headings list all functional entities and events necessary to completely describe the dependency structure of the design outline. Functional entities are identified in the headings by means of functional entity identifier codes. Events are identified by listing the event measurement point and, for critical events, the signal name.

10.4.1.3 Operation column. The operation column identifies all actions required to disclose the complete dependency structure at any desired level of coverage. Early in the life cycle, the operation column would consist primarily of mode identification. Later, as more detailed information is known, the operation column would take on more detail. In addition to mode identification, it would include other actions required to develop the dependency structure such as turn-on, check-out, and other unique procedures.

10.4.1.4 Event identifications and data. The event identification area of a design outline contains: (a) event descriptions, and (b) general notes. Each event occurring throughout the body of a design outline is assigned a unique identification number. This number appears directly below the event entry in the heading. The same identification number also appears in the event identification area. Here, the event is precisely specified by use of waveforms, timing diagrams, or other descriptive methods. In this manner complete performance characteristics associated with each event are identified. In addition, reliability and maintainability information is provided by means of data rows at the top of the design outline. Additional information can be presented simply by adding more data rows.

10.5 Supplemental information (as described in 20.12). There are specific types of data necessary to supplement the information provided by the design disclosures. This data falls into four general categories:

(a) System-equipment description
(b) Packaging data
(c) Specific disclosures supplemental data
(d) General supplemental data

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10.5.1 System equipment description. The system-equipment description provides management level description. This description includes an overall functional description, system effectiveness characteristics including overall performance characteristics, overall reliability and maintainability characteristics, man-machine interface, personnel requirements, physical features, and support equipment.

10.5.2 Packaging data. Packaging data is used to disclose both the overall and detailed physical design of the system-equipment. This data will include such items as detailed mechanical drawings, plan views of equipment layout, removal and replacement data and access requirements. Unless otherwise specified in the contract, these drawings shall meet the requirements of MIL-STD-100.

10.5.3 Specific supplemental data. There are times when data recorded directly on a particular design disclosure needs support data. This data is recorded on supplemental data sheets referenced on the disclosure. In addition, there are times when it is physically impossible to record all pertinent data directly on a given disclosure. In these cases, the most critical data is recorded on the disclosure and the remaining data referenced to supplemental data sheets.

10.5.4 General supplemental data. The general support data includes such items as cost estimates and expenditures, logistics data, parts data, alignment procedures, etc. Other data that is not recorded elsewhere but contributes to design disclosure should be recorded here. Where applicable, this data shall appear in a format specified in MIL-STD-100.

10.6 Life cycle adaptations. The intent of this handbook is to provide design disclosures which are compatible with the life cycle of the system or equipment being procured. Since the life cycle of all systems or equipments is not the same, the disclosures should be tailored to the individual needs of a specific procurement. In order to do this, the procuring agency must ensure that the review teams know the disclosures and general content that should be expected at each phase of the life cycle. With this basic knowledge, the review teams will find that the design disclosures greatly aid in the assessment and analysis of the system-equipment. At the same time, the contractor should be made aware of the extent of his commitments in preparing the disclosures. The following paragraphs describe briefly the identifiable outputs of each phase of a typical life cycle and the types of disclosures and general content which can be expected. Figure 6 provides a summary of the system-equipment life cycle and the adaptation and evolution of the design disclosures during each phase.

10.6.1 Concept formulation phase (see figure 6). The concept formulation phase is the initial program phase where system requirements are identified in sufficient detail to form a basis for the following phases. The development of a system during this phase begins with a General Operational Requirement (GOR) which identifies both the needs (Threat) and the activity (Mission) requiring a system. An analysis of the GOR results in Proposed Technical Approaches (PTA) which are then analyzed and evaluated against the feasibility criteria which also have been established. The feasibility evaluation of the PTA results in an approach which appears to meet the need. This evaluated approach then will result in a Specific Operation Requirement (SOR) and a Technical Development Plan (TDP) which in turn will allow contractual performance of the following life cycle phases.

At this time master block diagram/texts can be prepared to facilitate analysis of the proposed technical approaches. Each block in the master block diagram defines a major functional division of the system. Text within each block will describe intended operation, alternate schemes, input-output signal characteristics, and reliability and maintainability goals.

The master design outline can be used during this phase to depict the interdependencies of the functional divisions for each proposed approach. The design outline clearly portrays the interrelationships of the input-output flow between functions in a manner not possible on the text diagram. Together with the supplemental data, these two disclosures will clarify and solidify the elements of the system to be described by the SOR and TDP.

10.6.2 Contract definition phase (see figure 6). During the contract definition phase, a system will be defined in terms of its functions, and equipments. Alternative equipment schemes are developed and analyses performed to develop a feasibility model. The result is a system specification for design and development.

At this time, master block diagram/texts and master design outlines can be prepared to facilitate analysis of equipment schemes. Each major block in the master block diagram/text defines a major functional

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division. The text for each block identifies the modes of operation, the general action that must take place within the functional division, reliability, maintainability and human factors goals and significant characteristics of its inputs and outputs.

While the master block diagram/text illustrates, in block form, the functional divisions and the signal flow among the functional divisions, the master design outline illustrates, in symbolic chart form, the interdependencies of the functional divisions. It clearly portrays the interrelationships of the signals in a manner not possible on the text diagram. It emphasizes the differences in the signal flow from mode to mode and provides precise signal identifications (if known). With the functional dependency structure defined at the equipment level, a broad approach can be taken to the problems of reliability, maintainability, and other elements of Integrated Logistics Support (ILS).

During the contract definition phase, very little actual data will be known. The disclosures at this time will reflect estimates, proposed schemes, and desired event and signal identifications. This does not make the disclosures any less useful. At this stage in the design, a formalized, logical data presentation scheme is essential in order to adequately evaluate the several alternate designs. Once an alternate scheme is determined, the disclosures provide excellent documents for disclosing the data required for the preliminary design phase.

10.6.3 Preliminary design phase (see figure 6). The purpose of the preliminary design phase is to develop the first design plan. This plan describes the manner in which the system or equipment outputs are to be mechanized. Design is complete to the circuit element (stage) level. Reliability and maintainability analyses are performed.

The master block diagram/text and the master design outline are updated during this phase to reflect the considerable amount of detailed information which is now available. The contractor will probably find it necessary to completely redraw the diagram to make the drawings compatible with the preliminary detailed block diagrams and detailed design outlines prepared during this phase.

Detailed block diagrams can now be prepared for each function. Circuit element level detail can be defined and the basic mechanization scheme clearly shown. Details such as interconnection data, test points, terminal boards, and hardware design may be lacking. However, as much detail as is available should be shown.

Blocked text for the detailed block diagrams can be prepared. The text should discuss how various circuit elements and composites work together. Modes of operation are also discussed. Performance goals are specified, and predicted reliability, maintainability, and other ILS information is included.

Detailed design outlines are prepared from the information shown on the detailed block diagrams. Predicted reliability and maintainability data is included at the top of the design outline so that analyses such as mean-time-to-repair, sensor placement analysis, modularization analysis, etc., can be performed. Predicted data should be originated by the contractor in accordance with contract requirements as specified by the procuring agency.

10.6.4 Engineering design phase (see figure 6). The engineering design phase is the beginning of the period of intensive development and design. During this phase, a developmental model is fabricated and is used to evaluate the performance capabilities of the system or equipment. Detailed design is complete down to the part level.

The master block diagram/text and the master design outline are updated to reflect all changes and additions that have occurred since the preliminary design phase. The appearance of these diagrams will normally not change a great deal since they are not too sensitive to part or circuit element level changes.

Since detailed circuit information is now available, blocked schematics can be generated. A blocked schematic is generated for each designated assembly.

Blocked text is also prepared for the blocked schematics. In addition to providing a detailed description of each functional entity on the schematic, calculated repair time and failure rate data are included.

Because of the great amount of detailed circuit information generated during this phase, the detailed block diagram is normally redrawn to include such information as interconnection data, packaging data,

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test points, and front panel markings. However, much of this data may still not be available until the detailed design phase.

When a new detailed block diagram is prepared, its companion blocked text must also be redrawn. In addition to describing the functional operation of the circuits, new qualitative and quantitative reliability and maintainability information should be included.

New detailed design outlines will have to be generated during this phase because of the large amount of detailed information now available. New analyses can be performed which will now begin to show the actual effectiveness characteristics of the system or equipment.

10. 6. 5 Detailed design phase (see figure 6). During the detailed design phase, the results of the analyses of the developmental model are evaluated and improvements are made to obtain the required performance characteristics. Specific details are worked out down to the part level. Design and packaging details are finalized, and a service test model constructed.

The disclosures applicable to this phase are virtually identical to those brought about by engineering changes which result from design review.

10. 6. 6 Test and evaluation phase (see figure 6). During this phase, the service test model is subjected to operational and environmental tests required to determine that the system or equipment requirements have been met. At this time, the detailed design outlines disclose the complete performance, maintainability and reliability characteristics of the proposed system or equipment.

The disclosures can now be used as input for the preparation of technical manuals, and as training material for military operators and technicians. The disclosures also prove to be of value for logistics planning.

10. 6. 7 Production phase (see figure 6). During the production phase, minor changes will probably be made for production convenience. These changes will require minor revisions to the formats to keep them up to date. During this period, the disclosures prove valuable aids to generate and administer the equipment acceptance test plan.

10. 6. 8 Installation and use phase (see figure 6). During the installation and use phase, no changes in the disclosures are expected other than those required because of possible field modifications. The design disclosures now form a complete, concise, and accurate history of the system or equipment development.

Copies of the finalized disclosures can be given to contractors as part of the data package for secondary procurement. And finally, the design disclosures will prove to be extremely valuable to the contractor as a guide for future designs. For special requirements the design disclosures set shall be implemented through supplementary handbooks in addition to the general handbooks.

20. FUNDAMENTAL TECHNIQUES

20. 1 Introduction. This section provides a detailed description for the development of each of the types of design disclosure. The order of presentation corresponds to the implementation of the disclosure in the system-equipment life cycle. Section 30 provides detailed rules for developing certain disclosure structures. These methods will be referenced in the appropriate paragraphs of this section.

20. 1. 1 Master block diagram/text. The master block diagram/text provides an overall functional definition of a system or equipment. The purpose of the diagram is to describe the operation of each major functional division in relation to that of the other functional divisions such that the total operation of the equipment can be understood and to provide access to lower levels of design information.

20. 1. 1. 1 Layout techniques (see figures 3 and 4). The master block diagram/text consists of major functional divisions enclosed by light solid lines functionally connected by signal lines. The functional divisions are identified by descriptive names and codes. Text within each functional division includes a description of its operation in relation to other functional divisions in addition to its reliability, maintainability and human factors characteristics.

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20.1.1.2 Functional blocks. Each major block on the diagram represents a major functional division. If a functional division is complex, additional blocks may be added within the major block to represent sub-functional groupings.

Each block is identified by a descriptive name and an identifier code. Identifier codes are discussed in detail in 30.2.

20.1.1.3 Interconnecting. The blocks are functionally connected by signal lines. Each signal line has coded arrowheads and is identified with a descriptive name. A list of arrowhead types and their meaning can be found on figure 7. Rules covering special use and layout of signal flow lines are given in 30.3.

20.1.1.4 Input and output signals. All input and output signals are clearly identified and a complete and accurate discription of each is given.

20.1.1.5 Front panel controls. During appropriate design phases front panel controls should be shown if they enhance a functional understanding (see 30.4). However, they should not be added if the number of controls is large and would tend to complicate the drawing.

20.1.2 Text coverage. Each functional block on the master block diagram/text will contain a description of how the function contributes to the overall operation of the equipment. Modes of operation should be discussed if important. In addition, detailed reliability, maintainability and human factors information is discussed. The MTTR, MTBF, and failure rate of each function are recorded. Significant reliability and maintainability features should be pointed out. If a discussion of the reliability and maintainability features is lengthy, they can be referenced to supplemental sheets. The supplemental sheets may also be used to expand the discussion of performance, human factors, and personnel requirements.

20.2 Master design outline (see figure 5). The master design outline as mentioned in 10.4 provides, in symbolic form, the overall dependency structure of a system or equipment. In most cases, there are two levels of design outlines necessary to show the complete logical dependency structure of a system-equipment: the system-equipment level (master design outline) and the detailed level (detailed design outline). More complex systems or equipments may require intermediate level design outlines. The master design outline is generated first. Its content includes the dependency structure of all functional entities as they are defined at this time. Failure rates (λs) are alloted to these functional entities in accordance with good engineering judgement, reliability data sources and proven prediction techniques. If the maintenance philosophy is known at this time, repair times for the various areas of the system or equipment can be estimated using prediction techniques. If desired, these estimated repair times (ERTs) can be estimated for each functional entity represented by a dot on the master design outline. Reliability and maintainability predictions can be made using the alloted λs and ERTs.

As the design proceeds and more circuit details are known, the detailed design outline is generated. The detailed design outline is an expansion of the master design outline and is directly keyed to it. The detailed design outline is described in 20.9. At this time the first circuit designs, test point selections, etc. have been made. Thus, λs and ERTs can be more precisely estimated for each functional entity on both the master design outline and detailed design outline.

As the design proceeds further to the point where schematic information is known, new reliability and maintainability data becomes available. Hence, new reliability and maintainability predictions can be made. This process continues until the final design of the system or equipment is complete.

20.2.1 Master design outline structure (see figure 4). The master design outline displays the overall dependency structure of the system-equipment in terms of defined functional groupings and functions which together constitute the system-equipment. The master design outline is divided into four main areas: (a) headings, (b) signal identifications and data, (c) operation column, and (d) body.

20.2.1.1 Headings (see figure 5). Column headings list the name and location of all events and functional entities associated with the master design outline. Events include:

(a) Inputs from external equipment
(b) Internal events
(c) Output event (terminal events) such as a PPI display or recorded output.
See 30.7.1 for specific rules covering heading entries.

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20.2.1.2 Signal identifications and data (see figure 5). When the master design outline is first generated (detailed design outline not generated yet), all major events appear on the master design outline. However, when the design progresses to the point where the detailed design outline is generated, additional events appearing on the detailed design outline will have developed which do not appear on the master design outline. Since the master design outline is a summary of the detailed design outline, all events should be directly keyed between the two design outlines. To accomplish this, the signal identification number associated with each event on the detailed design outline must be used for that identical event on the master design outline. Because of the direct keying, the signal identification numbers on the master design outline may not be sequential from left to right, but will provide a direct reference to the appropriate detailed design outline. See 30.7.2 for details concerning signal identification number assignment.

There are three data rows just below the headings. They are:

(a) Signal identification Number
(b) Failure rate (λ)
(c) Estimated repair time (ERT)

These rows provide the three basic ingredients which constitute system effectiveness. The first row contains numbers which key events to signal identification normally listed at the right of the diagram. These signal identification provide precise performance data including timing and voltage data, waveform data, and human factors data where applicable. In the second row, failure rates of subfunctions, composites, etc. are recorded, thereby providing reliability data. In the third row, ERTs of the subfunctions, composites, etc. are recorded providing maintainability data. Additional information can be presented simply by adding more data rows.

It should be pointed out that the design disclosures are for the placement of data, and are not requirements for data. Data recorded on the design disclosures is to be prepared in accordance with contractor data requirements as specified by the procuring agency.

20.2.1.3 Operation column (see figure 5). The operation column must identify all modes of operation. It may also display selected procedural actions if so desired. If the design has proceeded to the point where the detailed design outline has been generated, the selected turn-on and check-out step numbers of the master design outline must correspond to those of the detailed design outline. Necessary actions by personnel to operate the system can be disclosed using work study techniques and keyed to the operation column. See 30.7.3 for details concerning the development of the operation column.

20.2.1.4 Body (see figure 5). The body of the design outline depicts the dependency structure of the system-equipment. The dependency structure uses the three basic symbols described in 10.4.1.1.1 to show the interrelationships between the functional groupings shown on the blocked text diagram and their input-output signals. The development of the dependency structure is determined by the procedural actions identified in the procedure column. See 30.7.3 for detailed rules covering the development of dependency structures.

20.3 Intermediate block diagram/text. When the functional divisions as defined on the master block diagram/text are sufficiently large, subfunctionalization is sometimes necessary as in the case of large systems. The intermediate block diagram text provides the transition between a major functional division on the master block diagram/text and the many detailed block diagrams which may be required to describe the functional division.

The intermediate block diagram/text is generated by dividing the functional division into discrete subfunctions. These subfunctions are interconnected in the same manner as on the master block diagram/text with complete signal nomenclature and characteristics. Text is provided within each subfunction block to describe how the subfunction contributes to the overall major function. Modes of operation should be discussed if important. In addition, detailed reliability, maintainability and human factors information is discussed. The MTTR, MTBF, and failure rate is also indicated.

Under rare circumstances the design complexity may dictate that the information normally presented on this disclosure format be separated into two disclosures - an intermediate block diagram and an intermediate block diagram text.

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20.4 Intermediate design outline. The intermediate design outline is a companion disclosure to the intermediate block diagram/text and is used to depict the complete dependency structure of the functional division illustrated on the associated intermediate block diagram/text. The construction of the intermediate design outline is approached in the same manner as the master design outline. As was the case in the intermediate block diagram/text, the intermediate design outline provides the transition between the master design outline and the many detailed design outline parts which may sometimes be required to describe the detailed functional dependency structure in complex systems.

20.5 Detailed block diagrams. The purpose of a detailed block diagram briefly described in 10.2 is to depict a major functional division using special symbology to show the interrelationship of the functional entities required to perform the function. In the early design phases, the detailed block diagram illustrates the proposed mechanization of a function. As the design progresses through the development life cycle, the detailed block diagram is updated, at milestone points, to reflect changes in circuit design and the incorporation of physical packaging. Each time the detailed block diagram is updated, more and more detail is added, until finally, the diagram contains the type of detailed information shown on figures 8 and 9.

The following paragraphs describe the detailed information to be included on the detailed block diagrams and the drafting techniques which are to be used to illustrate this information. Naturally, many of the paragraphs will not apply to the earlier phases simply because the information is not available. However, as the design progresses, each of the techniques discussed will be employed.

20.5.1 Functionalization. A detailed block diagram is prepared for each major function defined on the master block diagram/text. Each detailed block diagram is developed in a manner which makes the technical operation of the function obvious. In order to do this, care must be taken to ensure that the complete function is shown on a single page. If this is not possible, the function is subfunctionalized in such a manner that the first page of a multi-sheet detailed block diagram shows the overall function and each additional sheet, the detail of each of the subfunctions. Each subfunction is shown on the first page as a box with a descriptive name, a functional entity identifier code, and a reference to the subfunction detail diagram. All inputs and outputs of each of the subfunctions are shown on both the first page and on the detailed diagram of the subfunction. By using this technique the entire function can be shown on a single page with secondary detail on succeeding pages. For smaller systems or equipments, a detailed block diagram page may cover more than one function and possibly the entire system or equipment. See 30.1 for details concerning functional grouping.

20.5.1.1 Functionalization within an analog function (see figure 8). The lowest level functional entities on the detailed block diagrams, functional elements (emitter followers, amplifiers, etc.), are represented by special block symbology (see 20.5.3.1) and identified by a functional entity identifier code (See 30.2). Functional elements which work together to perform a specific task (I-F amplifier, pulse generator, etc.) are grouped together and called composites. Each composite grouping is enclosed by a light solid line, and given a name and identifier code. The composite provides a means of discussing functional element interrelationships and allows consolidation of reliability and maintainability data (failure rates, estimated repair times, etc.) associated with the functional elements.

20.5.1.2 Functionalization within a digital function (see figure 9). In digital systems or equipments, the detailed block diagram takes the form of the familiar logic diagram with functional grouping techniques applied. As in the case of analog detailed block diagrams, the logic diagram represents a complete function. Each of the logic operations being performed is represented by discrete logic symbols. These logic symbols are grouped into discrete functional entities. This functional grouping is analogous to the functional grouping of parts on blocked schematics: that is, grouping the minimum number of logic symbols required to perform a discrete functional operation. Each of the functional groupings is enclosed with a light solid line and given a name and a logic functional entity identifier code.

If additional functional groupings are necessary above this level, the compositing technique may be employed.

20.5.1.3 Wraparound schematic information (see figures 8 and 9). Blocked schematics are not prepared for cabinet wiring or other similar interconnecting hardware devices because complete interconnecting information appears on the detailed block diagrams and power distribution diagrams. However, cabinets and other wraparound devices frequently have components such as switches, relays, termination resistors, etc., mounted directly on the cabinet. Preparing blocked schematics of cabinet wiring in order to show

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these components is costly and does little more than increase the number of diagrams to be prepared. This can be eliminated by showing the components in schematic form on the detailed block diagrams. Wraparound circuit parts are therefore shown in schematic form on the detailed block diagram using the rules set forth for blocked schematics (see 20.10).

20.5.2 Layout (see figures 8 and 9). The layout of a detailed block diagram should be purely functional and should not be compromised in order to keep hardware of functional boundaries square or rectangular. Hardware boundaries and functional boundaries may be distorted as necessary to keep main single flow generally from left to right and feedback signal flow from right to left. The technical layout should never be distorted solely for artistic balance or to fill up white space.

20.5.3 Symbology.

20.5.3.1 Symbology in analog equipments (see figure 8). Blocked symbols including squares, rectangles, triangles and "truncated triangles" are used to designate various types of circuits and circuit elements. The definitions of the symbols are given in figure 7. During early design phases, these symbols can be used to show the proposed mechanization scheme of the functions. During later phases, when blocked schematics are available, the blocked symbols will be used to symbolize the functional entities defined on the blocked schematics.

20.5.3.2 Symbology in digital equipments (see figure 9). The logic block diagram of figure 9 is an example of the use of USAS Y32.14 symbology. The shape of the symbol serves to define the logic operation being performed. The presence or absence of the small circles at the inputs or outputs of a symbol indicate the relative voltage state necessary to activate the logic element and state of the resultant output. Tagging information around the periphery of the symbol identifies the input-output pin numbers. Tagging information inside the symbol identifies the logic element type and physical location. Note on figure 9 that the same logic element type may perform two different logic functions.

20.5.4 Functional entity identifier codes (see figures 8 and 9). Each functional element (except relays, switches, and single-part functional elements), composite, and function has an identifier code. The identifier code is used to precisely identify functional groupings in relation to their hardware or function location. See 30.2 for details concerning identifier code assignment.

20.5.5 Hardware boundaries (see figures 8 and 9). Heavy long dash-short dash lines are used to define hardware boundaries. The long dash is followed by short dashes, the number of which determines the level of hardware containment. A single dash indicates the overall system or equipment with two or more dashes indicating lower levels of hardware containment.

20.5.6 Hardware nomenclature and reference designations (see figures 8 and 9). Each assembly or subassembly which has a military nomenclature and reference designation, or other identifying number is identified as follows:

(a) Reference designations are printed in the upper left corner of the hardware boundary, followed by the official nomenclature. If official nomenclature is not assigned, colloquial nomenclature may be used.
(b) The words REF DESIG PREFIX, followed by the reference designation are printed in the lower right-hand corner of the hardware boundary.

20.5.7 Coded signal flow (see figures 8 and 9). All signal flow is coded by use of special arrowheads and appropriate annotations. Figure 7 lists the various types of signal flow codes used. These codes are not to be considered all-inclusive. If future needs require, new coded signal flow arrowheads should be generated.

20.5.8 Interconnections (see figures 8 and 9). All functional interconnections between hardware units are shown. Power connections are illustrated on the power distribution diagrams. All plugs, jacks, and cable numbers which provide the interconnection information between the units, assemblies, etc., are shown. Point-to-point wiring between circuit parts within an assembly need not be shown, but all interconnections between assemblies must be accounted for.

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20.5.9 Tie-in to other functions (see figures 8 and 9). For larger equipments, one detailed block diagram is prepared for each function. Normally, there are from five to ten functions associated with a system-equipment. The interconnection between these functions must be shown. To accomplish this, all signals entering or leaving the function associated with a particular detailed block diagram must be interconnected with functions associated with other detailed block diagrams. The input and output tie-in signals are terminated in the appropriate functional entities in their respective mating function. If available, all hardware and cabling information between functions must be shown. When tie-in is to another system or equipment, the signal identification and other pertinent interface considerations are included.

20.5.10 Special layout techniques (see figures 8 and 9).

20.5.10.1 Signal flow. Functional signal flow will be from left to right whenever possible. Feedback signal flow will generally be from right to left. Signal lines are connected to illustrate subordination and coordination of signal importance.

In general, coded signal lines are not used on logic block diagrams. However, since common sense must prevail, if a coded signal flow line will enhance understanding, by all means use it. All signal lines having significance must be labeled with a descriptive name and logic term. See 30.3 for details concerning flow layout.

20.5.10.2 Signal names and waveforms. Signals of importance should be identified by using signal name flags pointing to the line to be identified. Examples of signal name flags can be found on figures 8 and 9. Waveforms or timing information can be added at significant points if they will aid in the understanding of circuit operation.

20.5.10.3 Adjustments and controls. Adjustments and controls are illustrated to indicate the physical means of adjustment. The symbols and their use is described in 30.4.

20.5.10.4 Test points. Test points tied to the electrical connection between two functional entities are located in the space between the two blocked symbols. If the test point is connected in the middle of a circuit which is represented by a blocked symbol, the test point will be shown connected to the blocked symbol in the same manner as for adjustments. Front panel test points and their associated nomenclature will be enclosed by a solid light line.

20.5.10.5 Ground and return lines. Care must be taken when drawing the blocked symbols to show ground symbols in the same manner as they are shown on the blocked schematics. All grounds shown outside of the functional areas on the blocked schematic must be shown on a detailed block diagram. It follows that those grounds shown inside the functional areas on the blocked schematics must not be shown on a detailed block diagram. Refer to 30.6 for examples of when and when not to show ground symbols.

20.5.10.6 Mechanical connection break technique. When parts of a multiposition switch, potentiometer, or relay must be separated to preserve functional signal flow, a special technique is used to quickly identify the separated parts. Refer to 30.5 for a detailed discussion of the use of this technique.

20.5.11 Spares information (see figures 8 and 9). Spare relay contacts, switch contacts, connector pins, and terminals located on the wraparound hardware are listed in tabular form in the note area of the diagrams.

20.5.12 Engineering references (see figures 8 and 9). Engineering reference information is provided in the notes to identify the specific engineering reference documents used to generate the detailed block diagram. Engineering references for assemblies which have separate blocked schematics need not be repeated.

20.6 Detailed block diagram texts. The detailed block diagram text which was introduced in 10.3 provides a description of the signals and controls within the function(s) covered by the detailed block diagram. Through the use of intermediate functionalization levels (composites etc.), it describes those general functional characteristics and relationships which are not possible to discuss at the blocked schematic level of design disclosure.

20.6.1 Layout. The layout of the diagram is similar to the layout of the detailed block diagram. The layout differs only in that the functional areas may be enlarged to accommodate the text; however, the functional areas are maintained in relative position to the functional areas on the detailed block diagram.

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20.6.1.2 Text style and coverage. Text for the functional areas is written using third person, present tense, active voice, indicative mood. Sentences need not be complete; the subject is understood. Sentences normally begin with a verb and describe "what" the circuit does - eg. "generates 4V, 10 usec pulse with a PRF of 4KC". Articles may be dropped to conserve space. Text for digital circuits frequently includes waveforms, timing diagrams, and logic equations.

The failure rate of each functional element is recorded. If possible, this failure rate is recorded as part of the functional element text. When text is written at the composite level, failure rates for functional elements within the composite are recorded in tabular form within the functional area representing the composite. Total failure rates for composites and functions are recorded at the composite and function levels. A distinction should be made as to which functional elements contribute to functional operation, which are associated with testing, and which serve in power distribution.

When the packaging scheme is known, the estimated repair time (ERT) of each functional element is recorded. ERTs of higher levels of functionalization (composite, function, etc) are also recorded. The ERTs associated with these higher levels of functionalization are weighted means of their constituent functional entities.

The organization of the text on the blocked text diagram is as follows:

(a) Text is written for each functional area on the block diagram.
(b) When the functional areas are part of a higher order entity (composite, etc.), the text may describe the functional operation of the composite rather than the individual entities making up the composite.
(c) When hardware is identified and when an assembly is completely functional, the blocked text may be enclosed in the area representing the assembly.

Text for (a) relay contacts, (b) relay solenoids, and (c) switches is written as follows:

(a) Relay contacts. Text for relay contacts describes the specific function performed. When contacts are in parallel functionally and are grouped by a higher order functional area, the text describes the parallel operation at the higher order functional area.
(b) Relay solenoids. Text for relay solenoids describes the specific function performed by the solenoid. No mention of the energizing voltage need be made.
(c) Switches. Switches are treated in a manner similar to relay contacts.

20.6.1.3 Notes, spares, and engineering reference. General notes, flag notes, engineering references and spares information included on the detailed block diagram are not repeated on the text diagram.

20.7 Power distribution diagrams. During the early phases of design, very little information pertaining to power is available. At this time the power distribution diagram is quite simple. As more details become available, the power distribution diagram is expanded to incorporate them. Not until the packaging scheme is well established and the power requirements known, does the power distribution diagram take on significant detail. However, during these early phases, certain basic techniques should be employed. These basic techniques, discussed in 20.7.2 include such items as functional entity definition and identification, coded signal flow, brevity and completeness of descriptive material, etc.

During the latter phases of design, the power distribution diagram shows the complete generation and distribution of power down to the assembly level. The size and complexity of the system-equipment determines the scheme by which this information is shown. When possible, all of the information is shown on one diagram. Depending upon the packaging scheme, the power distribution diagram can be completely schematic in form or partially schematic and partially block diagram. When the complexity of the system-equipment is such that complete power generation and distribution cannot be shown on one diagram, a scheme must be developed to show this information in a coherent fashion. One suggested scheme is discussed in detail in the following paragraphs. Other equivalent schemes may be used if desired.

20.7.1 Power distribution scheme for large complex system-equipments. One prime power distribution diagram is generated to show the distribution of power from the prime power source to the various cabinets making up the system-equipment. Generation and distribution of power within the cabinets is shown by a separate cabinet distribution diagram for each cabinet.

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20.7.1.1 Prime power distribution diagram. The prime power distribution diagram shows the complete distribution and interlocking between system-equipment cabinets. All cable numbers, plug, pin, and jack numbers are provided. If there is a prime power cabinet, the generation and distribution of power within the cabinet is shown along with the distribution and interlocking to the other system-equipment cabinets. Power supplies and regulators packaged as units, and which have separate blocked schematics, are shown in block form. For these portions of the diagram, techniques for detailed block diagrams apply (see 20.5). All other portions of the prime power distribution diagram are shown schematically, and the rules for blocked schematics apply (see 20.10).

20.7.1.2 Cabinet power distribution diagrams. Cabinet power distribution diagrams show the generation and distribution of power to all assemblies within the cabinet. All cable numbers, plug, pin, and jack numbers are provided. Cabinet mounted parts are shown in schematic form. All power supplies and regulators, which are packaged as nomenclatured units, and which have separate blocked schematics are shown in blocked form.

20.7.2 Basic techniques (see figure 11). Figure 11 is an example of a power distribution diagram of a system-equipment in which power can be covered in one diagram. This power distribution diagram is shown completely in schematic form due to the simplicity of the power generation and distribution structure of the system-equipment it represents. Notice that techniques for blocked schematic construction have been employed.

20.7.2.1 Data presentation. Power generation is shown on the left of the power distribution diagram, signal flow is left to right, and power distribution is shown on the right of the diagram. A minimum of crossed lines and doglegs is desirable. Decoupling filters may be included on the power distribution diagram or may be located on the appropriate schematics depending upon the presentation emphasis desired.

Power distribution diagrams often contain an immense amount of data on a single page. For this amount of information to be on a page without sacrificing clarity, several techniques involved in the construction of the diagram need be employed. The first technique involves layout manipulation. By manipulating the layout, related circuit parts can be positioned to enhance understandability. Also, unnecessary long lines can be shortened to conserve valuable space. The second technique involves distorting the hardware boundaries, where necessary, to preserve smooth signal flow. Smooth signal flow is of prime importance when high density of data exists on a diagram.

20.7.2.2 Relays and switches. Each relay or switch contact set constitutes a functional entity. When relays or switches have more than one contact set, the contact sets are nomenclatured K1-A, K1-B, etc., beginning with the lowest numbered swinger. It is desirable to align the contact sets if possible, but not at the expense of functional flow. Relays and switches will not necessarily have all their contact sets operating in the same functional area. Often, they will function if different circuits. When this occurs, the various sets of contacts and actuating elements are connected by a dashed line to illustrate simultaneous operation. When the contacts are so far apart that the long dashed line clutters up the diagram, a simplifying technique can be used. One such technique is discussed in 30.5.

20.7.2.3 Grounding techniques. The grounding of a system-equipment is usually as complex as power distribution and, as such, requires equal attention. Refer to 30.6 for a description of proper grounding presentations.

20.7.3 Engineering reference. Engineering reference information is provided in the notes to identify the specific engineering reference documents used in the generation of the power distribution diagrams.

20.7.4 Spares information. Spare relay contacts, connector pins, terminal connections, switch contact sets, etc., are not shown on the body of the diagram but are listed in tabular form in the notes area of the illustration.

20.8 Power distribution diagram texts. The blocked text for a power distribution diagram provides a complete description of the generation and regulation of all power within a system-equipment. Its content and structure is identical to that for blocked schematic text in most areas. For those areas that are shown in block diagram form, its content and structure follow the rules for detailed block diagram text.

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Layout is identical to that of the power distribution diagram unless the text requires expansion of the functional entity boxes. If this is the case, the functional entity boxes may be enlarged or distorted slightly to accommodate text, but the relative positions of the boxes are maintained. General notes, flag notes, engineering reference and spares information included on the power distribution diagram are not repeated on the text diagram.

20.9 Detailed design outline (see figure 12). The detailed design outline is generated after the design process reaches the point at which most of the circuit details are known. It is an expansion of the master design outline (and intermediate design outlines as necessary) and is directly keyed to them. The detailed design outline discloses the dependency structure of a system-equipment in its most detailed form.

20.9.1 Detailed design outline structure (see figure 12). Each detailed design outline represents one or more complete functions. It displays the interrelationships among functional entities found on detailed block diagrams and is directly keyed to them. Each detailed design outline is divided into four main areas: (a) headings, (b) signal identifications and data (c) operation column, and (d) body.

20.9.1.1 Headings (see figure 12). Column headings list the signal name of all critical events and hardware identification of all events. The name, functional code, and hardware identification of functional entities associated with the detailed design outline are also listed. Events include such conditions as:

(a) Availability of signals or power inputs
(b) Indications that may be observed
(c) Conditions or states of the system or equipment - i.e. relay energizes, temperature normal, etc.

Functional entities include such items as:

(a) A portion of a part
(b) A part
(c) A functional element (group of parts)
(d) A composite (group of functional elements)
(e) A replaceable module, card, integrated circuit, etc.

The identifying codes of function entity entries are compatible with the coding on the detailed block diagram or the power distribution diagram Refer to 30.7 for detailed rules concerning the headings.

20.9.1.2 Signal identifications and data. There are three data rows just below the headings. They are:

(a) Signal identification number
(b) Failure rate (λ)
(c) Estimated repair time (ERT)

These rows provide basic performance, reliability, maintainability and human factors information. The function of these rows is the same as that described under master design outlines (see 20.2.1.2). Additional information can be presented simply by adding more rows. Refer to 30.7.2 for rules concerning event and signal identification number assignment.

It should be pointed out that the design disclosures are a format for the placement of data, not a requirement for data. Data recorded on the design disclosures is to be prepared in accordance with contractor data requirements as specified by the procuring agency.

20.9.1.3 Operation column (see figure 12). The operation column identifies all modes of operation. It also displays all operational actions which are necessary to provide complete logical design outlines. These actions consist of turn-on and check-out steps. These steps may take the form of making switch selections, providing external stimulus, or running diagnostic programs. Refer to 30.7.3 for development of operation column.

20.9.1.4 Body (see figure 12). The basic mapping rules involved with design outline construction are given in 30.7.4.

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20.10 Blocked schematics. In section 10.1 blocked schematics were shown to represent the most detailed level of design disclosure. Because of the detailed nature of the data required, blocked schematics are usually not prepared until detailed design data at the circuit part level is available.

When the packaging scheme of a system-equipment is known, one blocked schematic is generated for each unit assembly except wraparound assemblies, cabinets, or enclosures. Wraparound schematic data appears on either the detailed block diagram or on the power distribution diagram. For purposes of this handbook a wraparound is an enclosure and contiguous elements that remain after complete removal of assemblies, chassis, printed circuit cards, etc.

For digital system-equipments blocked schematics need only be generated for each of the unique circuits used as building blocks. Blocked schematics are also generated for all unique analog portions of digital equipment. The techniques for developing blocked schematics for digital circuits are the same as those for analog circuits.

When the packaging scheme of a system-equipment is not known, a blocked schematic consists of a group of related circuit parts that may perform one or more tasks. It is the prerogative of the procuring agency to define the exact areas of coverage of each schematic. However, if the procuring agency so desires, it can delegate this responsibility to the contractor.

20.10.1 Diagramming techniques (see figure 13).

20.10.1.1 Functional elements.

20.10.1.1.1 General definition. All circuit parts (resistors, capacitors, transistors, etc.) are grouped into functional entities (functional elements) and are enclosed by light solid lines.

Functional elements are defined as follows:

(a) Circuit parts which work together to perform a discrete functional operation are grouped to form function elements. Some examples shown on figure 13 are: X-CLP-1 (clamp), Q-EF-1 (emitter follower), and Q-MMV-1 (monostable multivibrator). It is also possible to have one functional element which is very closely related to, and which performs a subordinate function to another functional element, to be within the higher order functional element. An example is N-VD-1 on figure 13. Q-LIM-1 (limiter) is the higher order functional element. N-VD-1 (voltage divider) performs a subordinate function to Q-LIM-1 by providing adjustable collector voltage. Definitions of these functional element codes is found in 30.2.
(b) A single circuit part that performs a discrete functional operation is identified as a functional element. An example on figure 13 is filter capacitor C16.
(c) A portion of a circuit part that performs a discrete functional operation is identified as a functional element. Examples would be contact sets on multiple contact relays and switches.

20.10.1.1.2 Part arrangement. After the functional elements are defined, the circuit parts which make up these elements are arranged so that the functional elements appear in their most understandable form. A great deal of emphasis is placed on arranging the circuit parts to enhance functional understanding. One technique which simplifies the circuit design is to completely separate each power and ground connection. Common ground or return points and common power points should not be tied together merely to conserve symbols.

20.10.1.1.3 Graphic symbols. Standard symbols shall be used for all circuit parts as defined in the Military Standards referenced in Section 2 of this handbook. Special symbols may be used if approved by the procuring agency.

20.10.1.1.4 Part data. As much of the following circuit part data as is available is recorded for each part:

(a) Resistors - reference designator, value, wattage, and tolerance.
(b) Capacitors - reference designator, value, voltage, and tolerage.
(c) Coils, motors, and other wire wound devices - reference designator, value, tolerance, and resistance when applicable.
(d) Power transformer - reference designator and voltage and current ratings of windings.

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20. 10. 1. 2 Functional boundaries. Each functional element is contained within a functional boundary defined by a light solid line. The size and shape of the boundary depends on the amount of space required by the circuit design and the amount of space required to describe the functional element on the blocked text page. It is quite common to see a large functional area with only a few parts enclosed. A good example is X-GAT-1 on figure 13. The additional space was necessary to enclose all of the text for the functional element. Various levels of functionalization are usually required to show groups of functional elements which work together to perform discrete functional operations. When the function of a number of functional elements cannot be adequately described without grouping them, they are grouped together within an overall functional boundary and called a composite. An example of a composite is C-DET-1 on figure 13. Notice that C-DET-1, along with other functional elements, is further composited by C-PG-1.

20. 10. 1. 3 Functional entity identifier code. Each functional entity (except relays, switches, and single-part functional elements) has an identifier code. Examples of identifier codes are given on figure 7. This list of codes is not to be considered all-inclusive. New codes can be generated where future needs demand. Refer to 30. 2 for generation and use of identifier codes.

20. 10. 1. 4 Hardware boundaries. Heavy long dash - short dash lines are used to define hardware boundaries. The long dash is followed by short dashes the number of which determines the level of hardware containment. A single dash would indicate an assembly, with two dashes indicating a subassembly, etc.

Hardware boundaries may or may not be square or rectangular in shape. Primary concern is first given to functional circuit signal flow. Hardware boundaries may be distorted as necessary to insure a signal flow that is generally from left to right across the schematic.

20. 10. 1. 5 Hardware nomenclature and reference designations: Each assembly, card, etc., which has a military nomenclature and reference designation, or other identifying number is identified as follows:

(a) Reference designations are printed in the upper left corner of the hardware boundary followed by its official nomenclature. If official nomenclature is not assigned, colloquial nomenclature may be used.
(b) The words REF DESIGN PREFIX followed by the reference designation are printed in the lower right corner of the hardware boundary.
(c) Reference designations for multiple use assemblies may be identified by flag note. The flag note would specify each of their uses.

20. 10. 1. 6 Coded signal flow. All signal flow is coded by use of special arrowheads and appropriate annotations. Figure 7 lists the various types of signal flow codes used. These codes are not to be considered all-inclusive. If future needs require, new coded signal flow arrowheads should be generated.

20. 10. 1. 7 Layout techniques. In the paragraphs to follow, a number of special techniques are given which are designed to simplify layout and increase technical comprehension. In addition, many of the rules set forth in these paragraphs must be followed in order to simplify the task of preparing detailed block diagrams.

20. 10. 1. 7. 1 Direction of signal flow. Functional signal flow will be from left to right whenever possible. Feedback signal flow will generally be from right to left. Input signals to the assembly should appear on far left of the page. All output signals should appear on the far right of the page. Feedback signal inputs to the assembly should appear on the far right of the page.

20. 10. 1. 7. 2 Subordination and coordination. Signal lines are connected to illustrate subordination and coordination of signal importance. Refer to 30. 3 for signal flow line layout techniques.

20. 10. 1. 7. 3 Signal names and waveforms. All input and output signals are given descriptive names as shown on figure 13. Signals generated in the assembly which connect to output plugs but which are not used are labeled "not used". Signals of importance which are generated in the assembly may be identified within the assembly by using signal name flags pointing to the line to be identified. Waveforms at significant points throughout the blocked schematics may be added if needed for clarity.

20. 10. 1. 7. 4 Adjustment and controls. Adjustments and controls are illustrated to indicate the physical means of adjustment. Refer to 30. 4 for adjustment and control symbols and their use.

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20.10.1.7.5 Test points. Test points tied to the electrical connection between two functional entities are located in the space between the two functional areas.

The nomenclature or descriptive marking of front panel test jacks is enclosed with a solid light line.

20.10.1.7.6 Ground and return lines. Refer to 30.6 for use of ground and return lines.

20.10.1.7.7 Mechanical connection break technique. When parts of a multi-section switch, potentiometer, or relay must be broken up to preserve functional signal flow, a special technique is used to quickly identify the separated parts. Refer to 30.5 for discussion of the use of this technique.

20.10.1.7.8 Power circuits. Power and power filtering functional elements are separated from other functional elements. They are grouped into a composite power entity labeled (C-PWR-1) and drawn at the bottom of the blocked schematic. B+, chassis ground, and return lines are not shown connected to the functional entities by solid lines, but the connection is implied through the use of symbols for grounds and returns, and by break symbols for supplied power. The connection between a power line from the composite power entity to a destination in a functional entity is made by the break line technique. The power line is terminated outside the composite power entity, picked up outside the functional entity to which it is being supplied, and continued to the proper point within the functional entity. To facilitate making the connection between break symbols, these symbols should be aligned vertically and the voltage indicated at the break symbols.

20.10.1.8 Spares information. Spare relay contacts, connector pins, terminal connectors, switch contact sets, etc., which are available for future use, are not shown on the body of the drawing but listed in tabular form in the note area of the illustration.

20.10.1.9 Engineering reference. Engineering reference information, when available, is provided in the notes block on the blocked schematic to identify the specific engineering reference documents used to create the blocked schematic.

20.11 Schematic blocked texts. The schematic blocked text which was initially described in 10.3 is a duplicate diagram of the blocked schematic, with descriptive text replacing circuit part symbols in the functional entity areas. It is always located on a page facing the blocked schematic, which facilitates the matching up of corresponding functional entities between the two diagrams. One schematic blocked text page is provided for each blocked schematic. The techniques for developing blocked text for digital circuits is the same as those for analog circuits. However, in many cases (especially in digital circuitry) the building block schematic is sufficiently small as to allow the blocked text to be placed adjacent to the blocked schematic on the same page.

20.11.1 Diagramming techniques (see figure 14).

20.11.1.1 Layout. Layout for blocked text is identical to that of the associated blocked schematic. One method for preparing blocked text is to have a duplicate vellum of the blocked schematic made and then opaque out the circuit part symbols. Blocked text is then substituted for symbols.

20.11.1.2 Text coverage. The blocked text describes only circuit details. A description of where signals come from, or where they go, or what parts make up the circuit is not necessary.

Text should provide the following information about a functional entity:

(a) Statement of what the functional entity is
(b) A description of what function it performs
(c) Timing information
(d) Theory of operation (for unusual circuits)

Text within the lowest level functional entity (functional element) describes the operation of the circuit parts within that functional entity. Text within the higher levels (normally composites) describes the interaction between subfunctional entities. Functional element or part action is described relative to the functional or subfunctional entity of which they are a part. The failure rate (λ) and estimated repair time (ERT) of each functional entity is recorded.

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20.11.1.3 Sentence structure. When space is restrictive, sentences need not be complete. A clause frequently conveys the technical meaning adequately; however, sentences should not be avoided just to create an engineering jargon.

Descriptive sentences should be constructed using the following preferred rules:

(a) Third person
(b) Present tense
(c) Active voice
(d) Indicative mood

Generally, sentences begin with verbs with the subject understood.

20.11.1.4 Functional entity identifier and name. The functional entity identifier code on the blocked schematic text is identical to the code on the blocked schematic. In addition, the name of the functional entity is recorded on the blocked text diagram and located immediately to the right, or if space is available, directly under the identifier code.

20.11.1.5 Equivalent circuits. Since all circuits are drawn in their most familiar form, simplified diagrams are not used. However, on occasion, equivalent circuits may be used to support the text of unique circuits with complex interrelation of circuit parts.

20.11.1.6 Identical circuits. If the circuits on the same diagram are identical, the phrase "identical to" may be used rather than repetition of the text description. Similar circuits are discussed by using the phrase "similar to" or "identical to" and noting the exceptions. If the description of the functional element is very simple, it may prove to be just as easy to describe the circuit as to say "similar to". If such is the case, the circuit should be described.

20.11.1.7 Notes, spares, and engineering reference. General notes, flag notes, engineering references and spares information included on the blocked schematic are not repeated on the block text and are opaqued off the duplicate vellum.

20.12 Supplemental information (as mentioned in 10.5). Supporting information to the design disclosure is provided in supplemental data sheets. Part of this data is referenced to specific formats; the remainder is general support data applicable to the entire design disclosure package. Specific areas of supplemental information may be requested by the contracting officer. The disclosure format shall be in accordance with MIL-STD-100.

20.12.1 System-equipment description. The system-equipment description provides management level information. It provides an overall functional description of the system-equipment, discloses its overall system effectiveness characteristics, and presents functional and physical characteristics of the system-equipment. It contains in summary form data and information from lower level formats. The diagram is first prepared as a conceptual overall disclosure and is updated with time.

20.12.1.1 Overall description (see figure 15). An overall functional description of the system or equipment is provided on the system-equipment description page. The description identifies, generally, what the basic objectives of the system-equipment are and how they are satisfied. Significant characteristics of the major system-equipment outputs are discussed and unique features of the equipment summarized. If required, a description of how the system-equipment operates in relation to its environment may be included.

20.12.1.2 System effectiveness characteristics (see figure 15). Upon determination of the system effectiveness measures, preductions of these measures are recorded on the system-equipment description. Early in the life cycle parameter allocations are proposed. Later these values are preductions. Finally, they become values resulting from system tests which indicate the realized measure of design effectiveness.

20.12.1.2.1 Overall performance characteristics. Overall performance characteristics are listed and consist of a set of detailed standards for all inputs and outputs of the equipment. Other pertinent data such as normal operating temperatures, resistance to vibration, etc. is also given. A tabular form of presentation, such as that shown on figure 15 is desirable.

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20.12.1.2.2 Overall reliability characteristics. Overall reliability characteristics are listed and include predicted MTBFs for: (a) the equipment, (b) specific modes of operation, and (c) terminal events. Significant reliability features are pointed out such as redundancy where it exists, part reliability, etc. A tabular form of presentation (see figure 15) is desirable.

20.12.1.2.3 Overall maintainability characteristics. Overall maintainability characteristics are listed and include predicted MTTRs for: (a) the equipment, (b) specific modes of operation, and (c) terminal events. Significant maintainability features are listed such as maintenance philosophy (modular replacement or part replacement), type of check-out (automatic, semi-automatic, or manual), monitoring features, etc. Again, a tabular form of presentation is desirable.

20.12.1.2.4. Human factors characteristics. Overall human factors characteristics will be described and include such items as personnel and manning, personnel equipment, performance aids and other information called out in MIL-H-46855 by the contracting agency.

20.12.1.2.5 Physical features (see figure 15). Physical features such as overall dimensions, weight, etc. are given.

The physical packaging of the system or equipment is disclosed. Cabinets, drawers, racks, assemblies, cards, etc. are identified. If the size and complexity of the system or equipment dictates, several levels of hardware presentation may be used. During the earlier phases of design the form of presentation would include line drawings, isometrics, and sketches. Later, photographs should be used.

Where applicable, a facilities plan is included on the system-equipment description. Key equipments and other items of importance are identified.

Front panel controls and indicators are identified. Panel layout and human factors are disclosed. Such physical items as cooling units, cables, valves, etc. should be located.

20.12.1.2.6 Associated test equipment (see figure 15). The test equipment required to maintain the prime equipment is listed. Special test equipment supplied with the prime equipment should also be recorded.

20.12.1.2.7 Layout (see figure 15). There is no set layout associated with the system-equipment description. Visual balance and clarity of data presentation should dictate its structure. If required, more than one diagram page may be used to provide complete system-equipment coverage.

20.12.2 Supplemental data referenced to specific disclosures (see figure 16). There are times when data recorded directly on a particular disclosure needs support data. For instance, it is desirable to show the calculations (using approved maintainability prediction techniques) leading to the ERTs recorded on the detailed design outline. This data is recorded on supplemental data sheet. Also, there are times when it is physically impossible to record all pertinent data directly on a given disclosure. In these cases, the most critical data is recorded directly on the format and the remaining data referenced to supplemental data sheets.

Figure 16 shows the type of data that could appear on supplemental data sheets. Notice that reference is made to specific disclosures by designating the disclosure type and page number. If a functional entity is involved, its identifier code is given. The data associated with a particular disclosure should be organized in alpha-numeric order. Referenced disclosures should be organized by starting at the highest level disclosure (system-equipment description) and working toward the lowest level disclosure (blocked schematic).

Notice the supplemental data referenced to the time signal generator equipment of the system-equipment description. It consists of an equipment level reliability block diagram. A diagram such as this is frequently needed to support the system-equipment MTBF prediction. The reliability block diagram shown is small and represents a very simple system-equipment. For a complex system-equipment, a reliability block diagram such as this could require a complete page.

Notice the supplemental data referenced to F-NCG-1 of the master block diagram/text. A reliability block diagram is sometimes needed to support the MTBF prediction of a function. Also, reliability and maintainability characteristics associated with a functional entity, such as F-NCG-1, are frequently recorded on the supplemental data sheets if space does not permit on the master block diagram/text.

Figure 16 also shows the type of data that could be referenced to the master design outline. Calculations supporting the modal and terminal event MTBFs and MTTRs recorded on the master design outline are normally shown in the supplemental data sheets.

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A sample of the material that could be referenced to the detailed design outline is also shown on figure 16. ERTs associated with functional entities (dots) on the detailed design outline are recorded on that diagram. The calculations supporting this data would normally be recorded on the supplemental data sheets as shown on figure 16. ERT calculations such as these could require many supplemental sheets.

Support data to other format types would be presented and referenced in a manner similar to the above examples.

20.12.3 General supplemental data (see figure 16). In addition to supplemental data supporting specific disclosures, the supplemental data sheets contain general support data.

This general support data includes such items as parts location, alignment procedures; logistic data; cost; etc.

Parts are best located by means of photographs or line drawings. Front panel items should be emphasized. The data should be organized in a logical manner for easy reference.

A tabular form of data presentation for alignment procedures is desirable.

Other data that is not recorded on the disclosures, but contributing to design disclosure, should also be recorded in the supplemental data sheets. The format for supplemental data sheets is specified in MIL-STD-100.

20.12.4 Packaging data. Packaging data is used to disclose both the overall and detailed physical configuration of the system-equipment. This data will include such items as detailed mechanical drawings, plan views of equipment layout, removal and replacement data, access requirements, etc. MIL-STD-100 applies to these drawings.

20.13 Front matter. Figures 7 and 17 represent a suggested format for the front matter of a design disclosure package. The format may be changed as desired by the contractor. However, the following information must be included.

(a) Title
(b) Table of contents or functional index (whichever is more appropriate)
(c) Explanation of all special symbols used
(d) Explanation of all functional entity identifier codes used
(e) Listing of all reference designated hardware (if available)

Additional information may be added if desired. More than one page may be used if necessary.

30. Formating methods.

30.1 The following paragraphs describe the techniques and methods for the development of the diagram disclosures and the design outline disclosure. Each of the techniques and disclosures are referenced in Section 20 to provide a cross reference for applicability.

30.1.1 Functional grouping. Master block diagram/text, detailed block diagram, and blocked schematic diagram all require functional grouping of the entities shown on the diagram. It is the intent of these paragraphs to describe the various levels of functional grouping and their application.

30.1.1.1 Functional element (see figure 1). The functional element is the lowest level of functional grouping. At the blocked schematic level, it is represented by a grouping of the minimum number of parts required to perform a discrete function. Examples include amplifier stage, emitter follower, current limiting resistor, etc. Each functional element is enclosed within a light solid line block and identified by an identifier code or reference designator (as in the case of the single part functional element). At the detailed block diagram level the functional elements are represented by special symbols and identified by means of the identifier code or reference designator. In some instances, it is necessary to provide some subfunctionalization within a functional element. On the blocked schematic diagram, this is accomplished by grouping the desired parts within a light line block and identifying them with an identifier code (see Q-LIM-1, figure 1). On the detailed block diagram, this subfunctional grouping will not appear since the primary functional element is represented by a symbol.

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30.1.1.2 Composites (see figures 1 and 2). The composite functional grouping consists of two or more functional elements which perform a functional operation. At the blocked schematic level, the composite is identified by a light solid line block surrounding the desired functional elements and an identifier code and name. At the detailed block diagram level, the composites defined at the block schematic level are defined in the same manner on the detailed block diagram. In addition, it is sometimes necessary to create a second level composite grouping on the detailed block diagram which consists of grouping one or more first level composites plus functional element symbols (see C-TCC-1, figure 2). This level of composite is identified in the same manner as the first level composite.

30.1.1.3 Functions (see figure 2). Functions represent the highest level of functional grouping and consists of any combination of the lower level functional groupings. Normally the function is represented only on the master block diagram/text. However, in the case of small system-equipments more than one function may be represented on a detailed block diagram. Each function is identified by means of a light solid line block, name and identifier code.

30.2 Functional entity identifier codes. Each functional entity (except relays, switches, and single-part circuit elements), composite, and function has an identifier code. The code is divided into three parts with each part separated by a hyphen. An explanation of each of the three parts is as follows:

(a) The first part is an alpha digit which defines the category:

Function level category

F - function

Composite level category

C - composite
L - logic (used only on logic diagrams)

Functional element level category

H - hydraulic
M - mechanical
N - network
Q - transistor
U - microwave
V - vacuum tube
X - diode

(b) The second part is a group of alpha digits which define the type of circuit:

VD - voltage divider
OSC - oscillator
NAND - nand gate
etc.

(c) The third part is composed of numeric digits which differentiate between similar types of functional entities used more than once in a function. Several oscillators within one function would appear as: Q-OSC-1, Q-OSC-2, etc. The numeric digit also provides unique identification of similar functions used more than once within an equipment. For example, the codes for clock channel A, B, and C within a signal timing generator equipment would be: F-CC-1, F-CC-2, and F-CC-3.

A list of identifier codes is given on figure 7. This list of codes is not to be considered all-inclusive and is provided as a guide. New codes (both category and circuit type) can be generated whenever the need demands. The lowest level functional entities (functional elements) are identified by code or reference designator only. Single-part functional elements do not carry a code and are identified by reference designations only. Composites and functions, however, should be identified by both an identifier code and a descriptive name. The complete alphanumeric code, which will place a functional entity in the system hierarchy

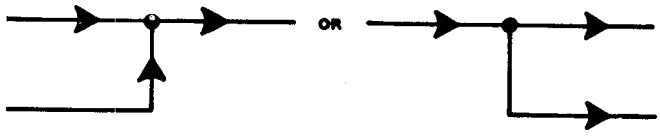
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structure and be compatible for computer processing, should be done on a per system basis with joint participation by procuring agency and contractor alike.

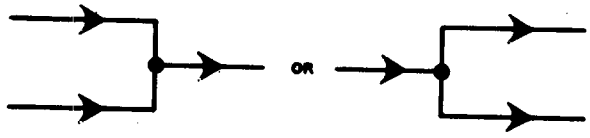
30.3 Signal flow lines. Functional entities are connected together by means of lines which show the signal or energy flow through the system-equipment. The following are a few general techniques which can be applied to make the signal flow through a system-equipment more obvious.

Functional signal flow will be from left to right whenever possible. Feedback signal flow will generally be from right to left. Signal lines are connected to illustrate subordination and coordination of signal importance.

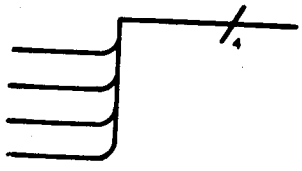
Subordinate junctions are those junctions in which one signal is less important than another. The less important signal joins the more important signal vertically. Subordinate junctions are shown as follows:



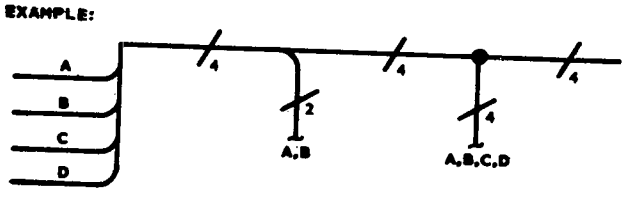
Coordinate junctions are those junctions in which all signals are of equal importance. Coordinate junctions are shown as follows:



Signal lines that contain closely related information may be grouped into one signal line. However, care must be exercised in not grouping signal lines indiscriminately. Signal lines are grouped as follows:



The number of signals being grouped is indicated by means of a slash through the grouped line and a number as shown above. Distribution of the grouped signals is shown by means of a curved breakoff line or a straight line breakoff.



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In the example, the first distribution uses the curved breakoff indicating only a portion of the grouped signals are being used. The second distribution uses a straight line breakoff indicating all of the grouped signals are used.

30.4 Adjustments and controls. Adjustments and controls are illustrated to indicate the physical means of adjustment. The symbols to be used are shown below:

Screwdrive adjust

Thumb adjust

Hexnut adjust

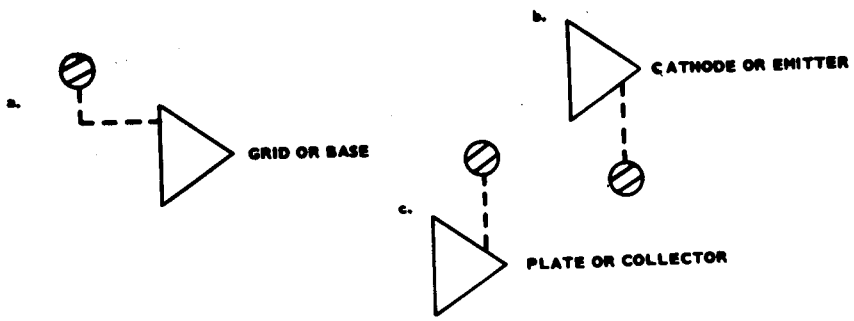
Knob switch

Toggle switch

Push-pull switch

etc.

If the adjustment is a front panel adjustment (can be seen with the equipment mounted, doors closed, and dust cover on), the adjustment symbol and adjustment nomenclature will be enclosed with a solid light line block. If possible, on detailed blocked diagrams, the adjustment symbol should indicate the relative circuit location of the adjustment as follows:



30.5 Mechanical connection break technique. When a multi-section switch potentiometer, or relay must be broken up to preserve functional signal flow, a special technique has been devised to quickly locate the various parts. At the bottom of figure 1, notice two black triangles with R49 printed in white. Directly above these triangles are two identical triangles mechanically connected to sections of potentiometer R49. The black triangles indicate a mechanical break. The white lettering indicates the reference designator of the part. On a complex drawing, other sections of a multi-section part can be quickly found by dropping to the bottom of a page and locating other black triangles with the same reference designator.

A refinement to this technique is to point all triangles at the bottom of the page toward the triangle at the bottom of the page which is directly below the relay solenoid or the section of a switch or potentiometer which shows the control knob and panel nomenclature. The triangle directly below the solenoid or control knob points up. All triangles within the diagram point down. By following the path pointed out by the triangles, the user can quickly locate the controlling element.

30.6 Ground and return lines. Symbols associated with specific types of return lines are as follows:

R — All D-C return connections

N — Unregulated A-C neutral connections

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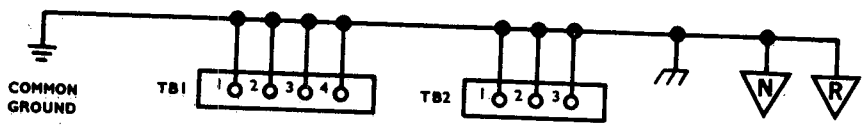


Regulated A-C neutral connections



Chassis ground

The individual grounds are handled and identified separately throughout the power distribution diagrams until they eventually are tied together at a common point. An example of how the tie point is shown is as follows:

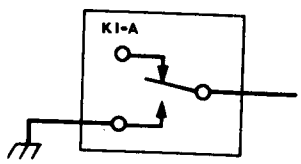


At the detailed block diagram level the ground symbol is placed outside the functional area it is associated with when it is required for functional understanding.

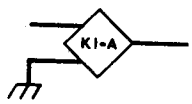
Examples:

1. Relay contacts

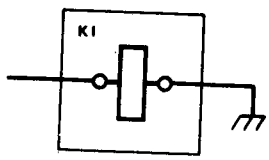
Blocked schematic

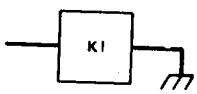


Block diagram

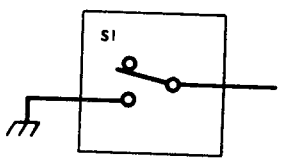


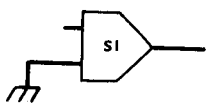
2. Relay solenoid





3. Switches



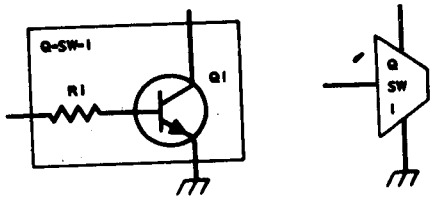


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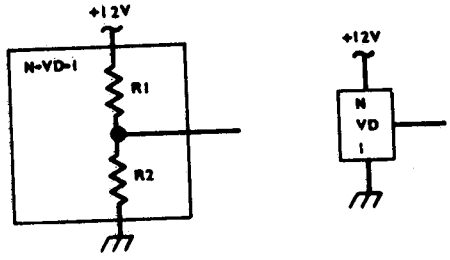
Examples (con't)

Blocked schematic Blocked diagram

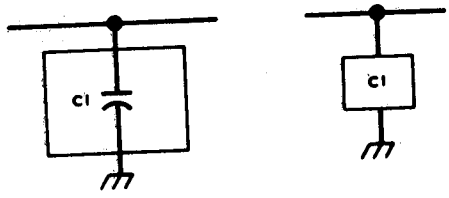
4. Transistor switches



5. Voltage dividers



6. Shunt filters

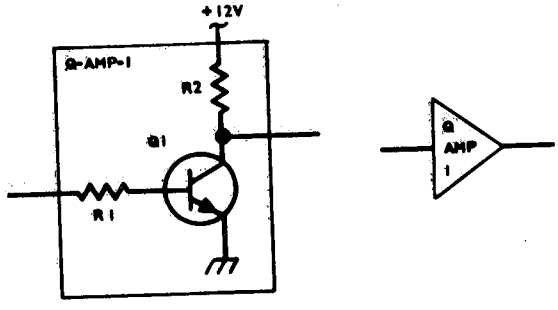


At the detailed block diagram level, the ground symbol is placed inside the functional area it is associated with when it is not required for functional understanding.

Examples:

Blocked schematic Block diagram

1. Amplifiers

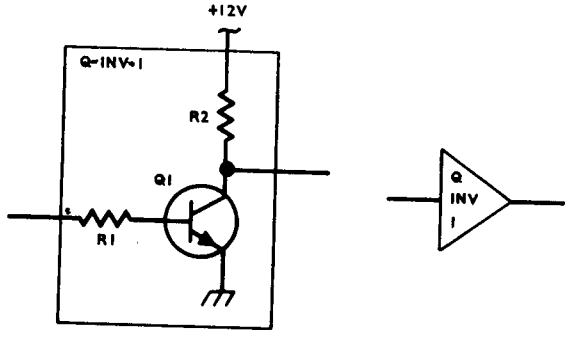


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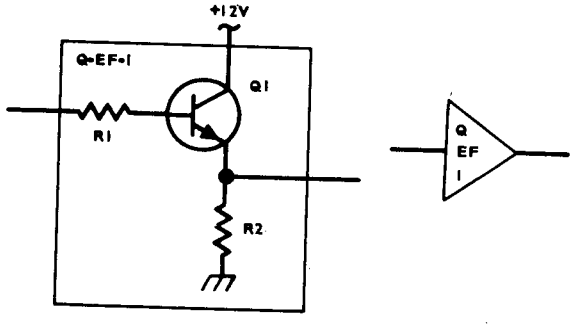
Examples (Con't.)

2. Inverters

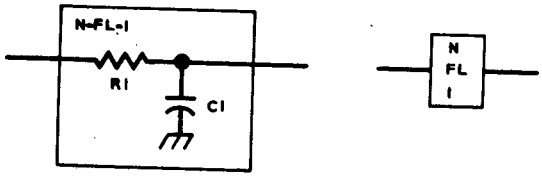
Blocked schematic Block diagram



3. Emitter followers



4. Filters

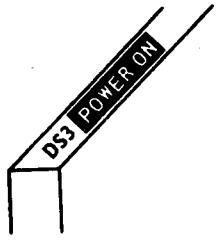


30.7 Design outline development. The following paragraphs describe the techniques and methods for the development of each of the parts of a design outline. The general description of the design outline is given in 10.4.

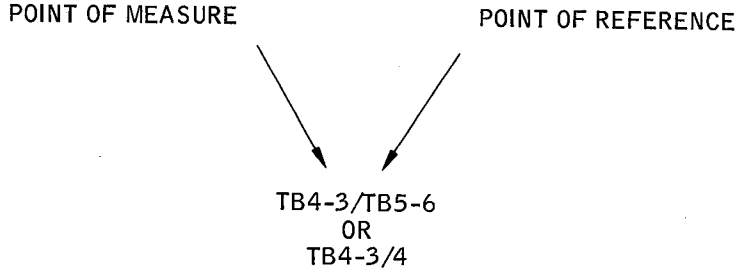
30.7.1 Heading entries (see figures 5 and 12). The heading of a design outline lists all of the events and functional entities to be depicted in the body of the design outline. The following paragraphs describe the basic rules for making heading entries.

30.7.1.1 Event entries. For events which are to be observed, the point of observation is illustrated as follows:

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When the indication is observable from the outside of the system-equipment, the panel nomenclature for the indication is illustrated by white lettering on a black background. For events which are to be measured, the points of measure are listed as follows:



The physical measurement points of all events are identified except during early design phases in which only the signal name is used due to lack of hardware information. In later phases, critical events are identified by listing first their signal names followed by, in parentheses, their physical measurement points. All other events are identified using only the measurement point. Identifications or descriptions for the event are referenced by a number located in the box created by the intersection of the signal identification row and column headings. When adjacent columns refer to the same event, the event entry can be so placed that it applies to more than one column.

30.7.1.2 Functional entity entries. Functional element entries are identified by means of their identifier codes. Composite and higher level functional entities are entered using both the functional name followed by the identifier code.

30.7.1.3 Heading entry hardware and function identification. The hardware identification of each functional entity and event is specified at the top of each column. Assigned reference designations are to be used for system-equipment identification. When adjacent columns refer to items in the same assembly, the location identifier can be placed so that it applies to more than one column.

In addition to the hardware identification, the master design outline heading entries must also reference the associated function. When adjacent columns refer to items in the same function, the function identifier can be placed so that it applies to more than one column.

30.7.2 Event and signal number assignment (see figures 5 and 12). All events requiring specific data are assigned a signal identification number. The identification numbers provide a reference to the detailed signal data normally located to the right of the design outline and also serve as a means of access between design outlines.

During early design phases when both intermediate and master design outlines are required, identification numbers are assigned sequentially at the intermediate design outline level. The identification numbers are then assigned to the corresponding events on the master design outline.

During the later phases when detailed design outlines are introduced, each detailed design outline is assigned a number. Signal identification numbers are then assigned, in numerical sequence, to each event on the design outline and are prefaced with the assigned number. For example, detailed design outline 1 signal identification number assignments are 1-1, 1-2, 1-3, etc. Detailed design outline 2 assignments are 2-1, 2-2, etc. Distribution event identification numbers are identified by white numbers on a black

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background. Distribution events re-entered on other detailed design outlines carry the originating identification number.

At this point in time, the intermediate and master design outlines must be revised to include the corresponding event identification numbers assigned at the detailed design outline level.

30.7.3 Operation column (see figure 12). The operation column specifies all of the man-machine actions required to completely exercise the system/equipment. The dependency structure within the body of the design outline is developed according to the procedures specified in the operation column. The following paragraphs describe the basic composition of the procedure column and the general sequence in which the procedures are presented.

30.7.3.1 Composition. For purposes of identification the procedures are divided into two basic categories: modal and test. Since the modal procedures are of primary concern these are presented first with the test procedures indented within the modal procedures. The dependency structure associated with each mode and test is identified by means of heavy horizontal lines (see figure 12). The modal procedures are numbered sequentially and the test procedures are assigned capital letters (A, B, C, etc.) sequentially within each modal procedure.

30.7.3.2 Development. The general procedure for the development of the procedure column is as follows:

(a) Determine what the different system/equipment modes are and what switch selections are necessary to obtain them.
(b) Present power turn-on as the first modal procedure. The dependency structure developed in the design outline body will reflect application of power, activation of power supplies and power distribution. Remember to include any test procedures as indented steps (see figure 12).
(c) Next, select the primary mode of operation and develop the complete dependency structure for this mode. Include all steps necessary to exercise all controls relative to this mode either as secondary modal steps or indented test.
(d) Select remaining modes in a logical dependency order and develop dependency structure as in (c).

30.7.3.3 Continuity scheme. In order to maintain continuity between the procedural steps on the master design outline and the many detailed design outlines, the modal and test procedure number assignments made on the master design outline are maintained on the detailed design outlines. This means that the modal step numbers for a given detailed design outline may not be sequential but will reflect the modes in which the depicted function is involved.

30.7.4 Body mapping. The body of a design outline is a mapping of a block diagram design. The mapping structure consists of basic symbols used to relate a functional entity or a group of functional entities to their associated input and output events. The following paragraphs describe the basic symbology and the mapping of block diagram designs.

30.7.4.1 Basic symbols. The three basic symbols used in mapping are defined as follows:

Event Box: ☐ to represent an action or availability of one or more events resulting from the proper operation of the functional entities associated with the event.

Functional Entity Dot: ● to represent a functional entity or a group of functional entities.

Dependency Marker: ▲ to indicate dependency upon another event.

Three basic types of event boxes are used in mapping. These are; (a) the single event box to represent one unique event, (b) the composite event box which consists of two or more single event boxes to represent multiple events and (c) the contracted event box which consists of a single event box to represent multiple events. Each of these event box types is discussed in more detail in the following paragraphs. Nomenclature in the event box (☐) specifies the type of action or availability of the event. Figure 7 lists the various nomenclatures that can appear in the box and their meaning. To assist in determining the accessibility of

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the events indicated on the design outlines, three distinguishable kinds of backgrounds are used within the event boxes. These backgrounds and their use are:

BLACK (with white lettering)	A	Events which may be recognized from outside the equipment. Examples are: Front panel meters, front panel lamps, PPI display, motor running, etc. Notice that this recognition includes events other than those which may be observed by sight.
GRAY (with black lettering)	A	Events which may be determined or observed only after the equipment enclosure has been opened, and which are readily accessible.
WHITE	A	Events which are difficult to access.

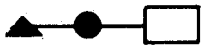
There are several variations of the functional entity dot. One such variation is the partial dot (◐) which indicates that the functional entity which it represents is only partially involved in producing the corresponding event. Other variations of the functional entity dot are described on figure 7.

There are also several variations of the dependency marker which are used to describe the relationship between a group of events and an ensuing dependent event. These variations and their use are described in later paragraphs.

30.7.4.2 Basic dependency structure. A dependency line consists of dependency markers, functional entities, dots and event boxes connected as follows:



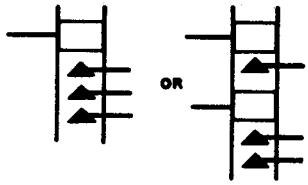
A dependency structure consists of a series of dependency lines. Each dependency line is constructed using the three basic symbols. The simplest dependency line would appear as follows:



By itself this dependency line has very little meaning except to indicate that the event (▲) is dependent upon the proper operation of the functional entity (●) and the availability of some previous event (□).

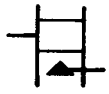
In order to provide meaning to this dependency line the symbols are ordered into one of two types of columns: the event/dependency column and the functional entity column.

The event/dependency column consists of event boxes and dependency markers depicted as follows:

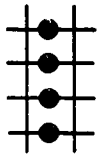


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The simplest event/dependency column would appear as follows:



The functional entity column consists of only functional entities, even though these entities may be depicted more than once. It is indicated as follows:



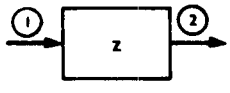
The simplest form of a functional entity column would appear as follows:



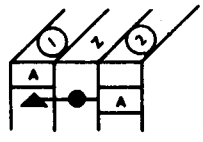
30.7.4.3 Mapping methods and techniques. Experience indicates that the following methods and techniques govern substantially all the basic mapping situations to be encountered. If situations arise which are not covered in the following paragraphs, the contractor should carefully augment the mapping schemes and record such for future use.

30.7.4.3.1 Simple dependency line. The mapping of a functional entity with one input and a single output results in a simple dependency line as follows:

BLOCK DIAGRAM



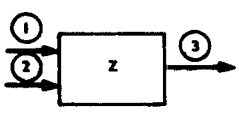
DESIGN OUTLINE



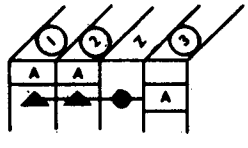
The mapping shows that the output event at ② is dependent upon a functional entity Z and the input event at ①.

30.7.4.3.2 Multiple input dependency line. A functional entity with more than one input and a single output is mapped as a multiple input dependency line as follows:

BLOCK DIAGRAM



DESIGN OUTLINE

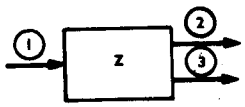


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Check the source to verify that this is the current version before use.

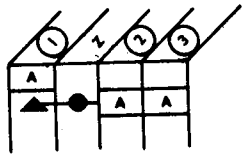
The mapping shows that the output event at ③ is dependent upon functional entity Z and the input events at ① and ② .

30.7.4.3.3 Multiple output dependency line. A functional entity having one input and multiple outputs is mapped as a multiple output dependency line as follows:

BLOCK DIAGRAM



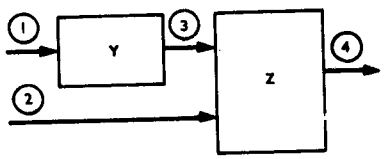
DESIGN OUTLINE



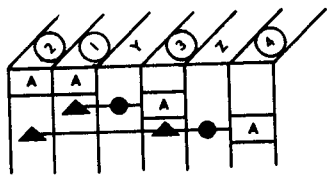
The mapping shows that both outputs at ② and ③ are dependent upon functional entity Z and the input event at ① .

30.7.4.3.4 Dependency chain. A dependency chain consists of a series of interrelated dependency lines. The following example shows a dependency chain consisting of two dependency lines.

BLOCK DIAGRAM



DESIGN OUTLINE



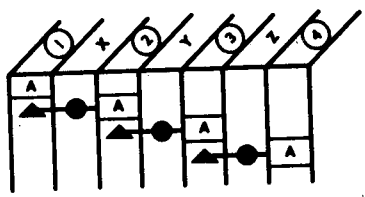
The last dependency line in this example shows that the output event at ④ is dependent upon functional entity Z and the events at ③ and ② .

30.7.4.3.4.1 Serial relationship. As a rule serial relationships are shown as follows:

BLOCK DIAGRAM

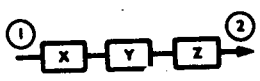


DESIGN OUTLINE

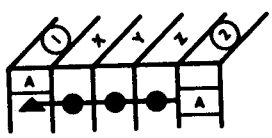


OR

BLOCK DIAGRAM

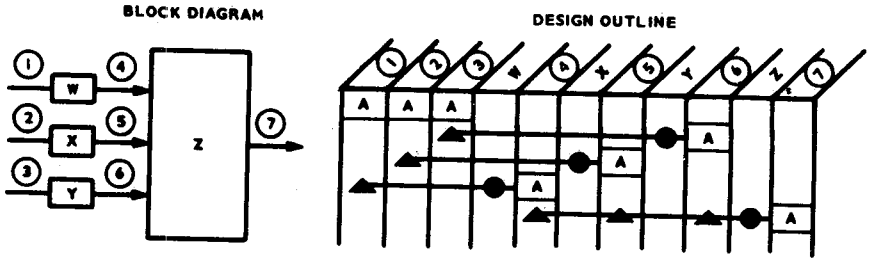


DESIGN OUTLINE

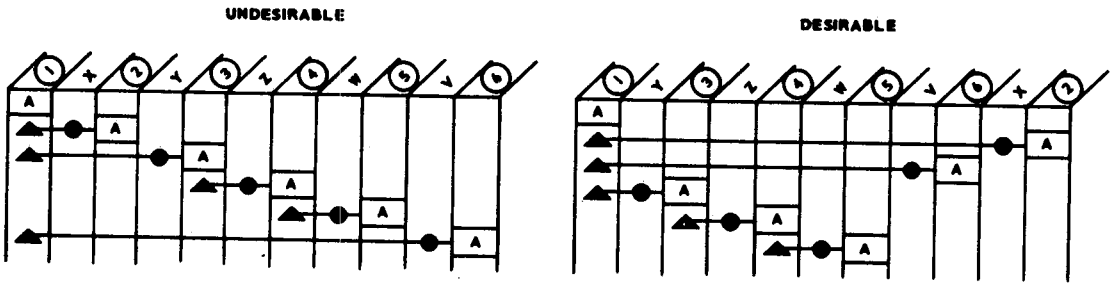


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30.7.4.3.4.2 Parallel relationships. As a rule parallel relationships are shown as follows:

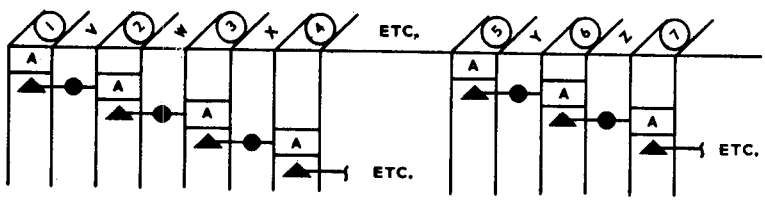


30.7.4.3.4.3 Serial parallel relationships. The following example illustrates the preferred manner in mapping serial parallel relationships:



The undesirable presentation erroneously infers by its structural layout that there is one serial dependency chain. The desirable presentation shows clearly by its structural layout that there are three separate parallel dependency chains. It also shows that one of the dependency chains has several serial relationships.

30.7.4.3.4.4 Horizontal line usage. Normally, only one dependency line is used per horizontal segment of the chart. However, if height is a problem on a design outline, more than one dependency line may be used if care is exercised. The dependency chains should be separated by ample space to avoid confusion. An example of this technique is as follows:

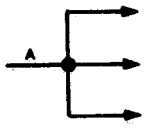


Notice that each dependency chain is easily identifiable.

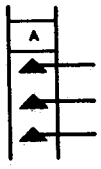
30.7.4.3.4.5 Parallel divergent branches. When a path diverges into a number of paths, it is depicted by an event/dependency column using a single event and the appropriate number of dependency markers. This situation is shown as follows:

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BLOCK DIAGRAM

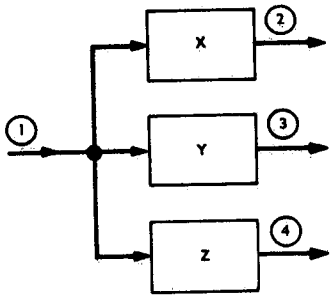


DESIGN OUTLINE

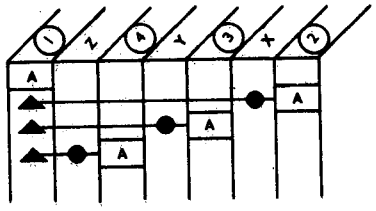


The following example utilizes this method:

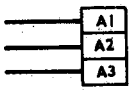
BLOCK DIAGRAM



DESIGN OUTLINE

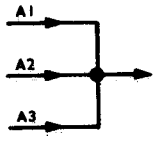


30.7.4.3.5 Composite events. A composite event represents an event which is made up of a number of single events. The constituent events may or may not be measured individually. A composite event is constructed in an event/dependency column as follows:

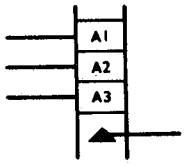


30.7.4.3.5.1 Parallel convergent branches. When paths converge into a single path, they are depicted by an event/dependency column using a composite event and a single dependency marker. This situation is shown as follows:

BLOCK DIAGRAM



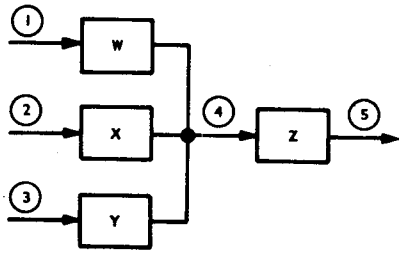
DESIGN OUTLINE



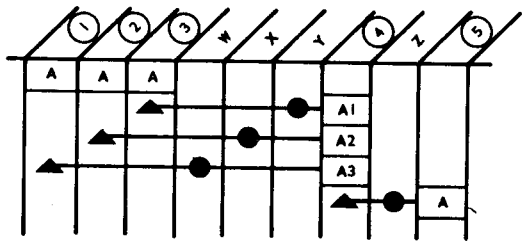
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:38Z
Check the source to verify that this is the current version before use.

The following example illustrates this method:

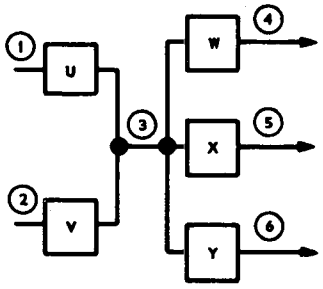
BLOCK DIAGRAM

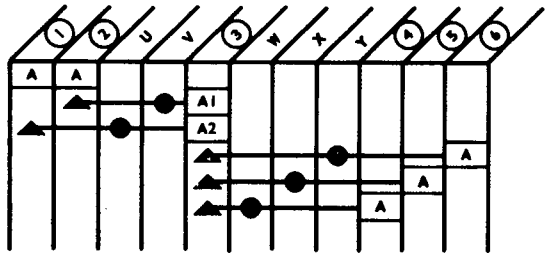


DESIGN OUTLINE



30.7.4.3.5.2 Convergent divergent branches. The following example illustrates a situation in which both divergent and convergent branches exist.



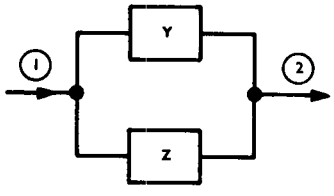


30.7.4.3.5.3 Redundancy. Redundancy is mapped using the composite event and a special dependency marker. The following examples illustrate the mapping of various types of redundancy.

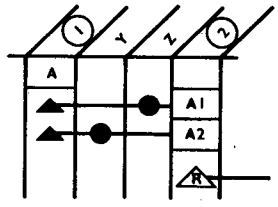
ACTIVE REDUNDANCY

Active redundancy is mapped as follows:

BLOCK DIAGRAM



DESIGN OUTLINE

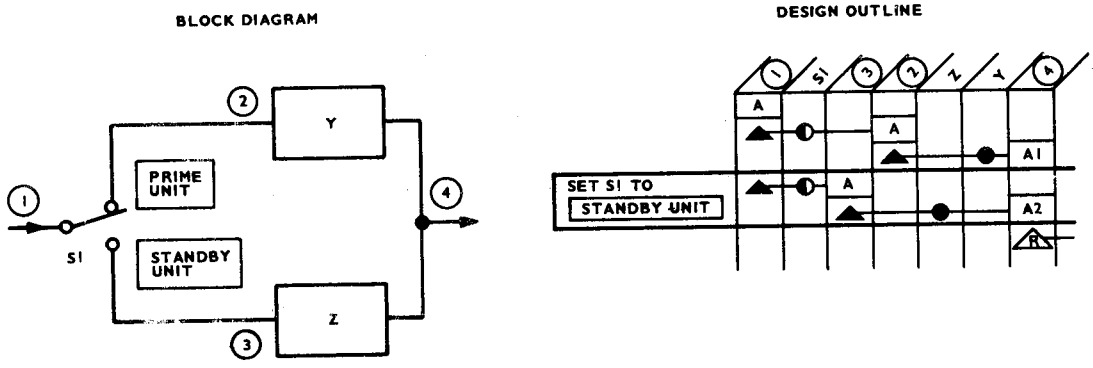


The design outline shows that if the signal at ① is available and either functional entity Y or Z is operating the signal at ② will be available. The notation R within the dependency marker denotes a redundant event.

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MANUALLY SWITCHED REDUNDANCY

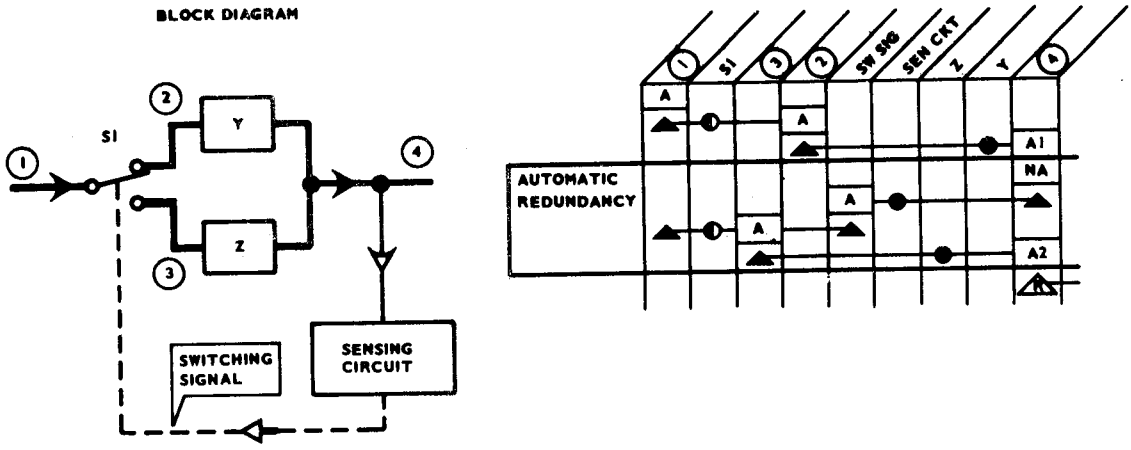
Manually switched redundancy is mapped as follows:



This situation is mapped in the same manner as the active redundancy situation except that the man operation is specified in the procedure column and outlined with a heavy line along with the associated dependency line.

AUTOMATICALLY SWITCHED REDUNDANCY

Automatically switched redundancy is mapped as follows:

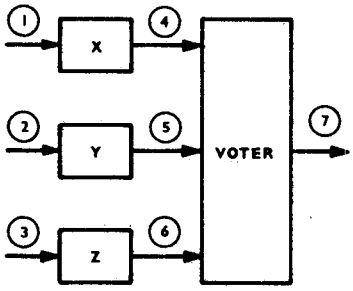


The design outline first maps the primary dependency structure. The automatic switching is emphasized by the heavy lines and the notation within the procedure column. The redundant signal is initiated by the use of a signal not available event box (NA) at ④ which generates a switching signal to activate the redundant path.

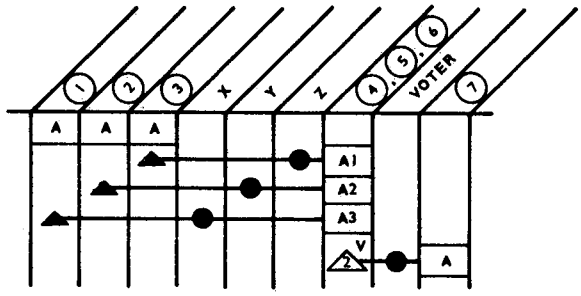
30.7.4.3.5.4 Voting. The following example illustrates the mapping of a typical voting situation.

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Check the source to verify that this is the current version before use.

BLOCK DIAGRAM



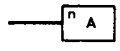
DESIGN OUTLINE



This example illustrates a 2/3 voter. Parallel events at ④, ⑤, and ⑥ are shown as a composite event. The dependency marker under the composite event has special notation as follows: The V denotes that the events are being voted. The 2 within the dependency marker indicates that any two of the three events represented by the composite event are required for the event at ⑦ to occur.

30.7.4.3.6 Contractions. In the event space on the design outline becomes limited, several schemes have been devised to allow contractions of events and functional entities. The following paragraphs describe the techniques for contractions. It is important to note here that contractions should be used only when absolutely necessary since they tend to obscure the dependency structure details.

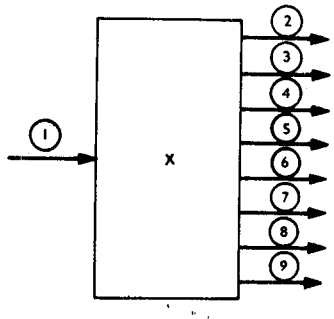
30.7.4.3.6.1 Contracted event. A contracted event is constructed in an event/dependency column as follows:



The n in the upper left hand corner of the contracted event indicates the number of events it represents.

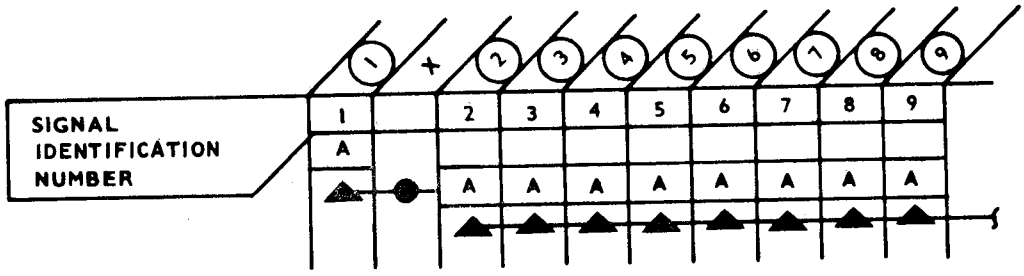
The following is an example of contracted events. Consider a functional entity X with one input and eight outputs. If all eight outputs are required as a unit by many different functional entities on the design outline, it is desirable to contract these outputs into one event entry.

The block diagram of functional entity X would appear as follows:



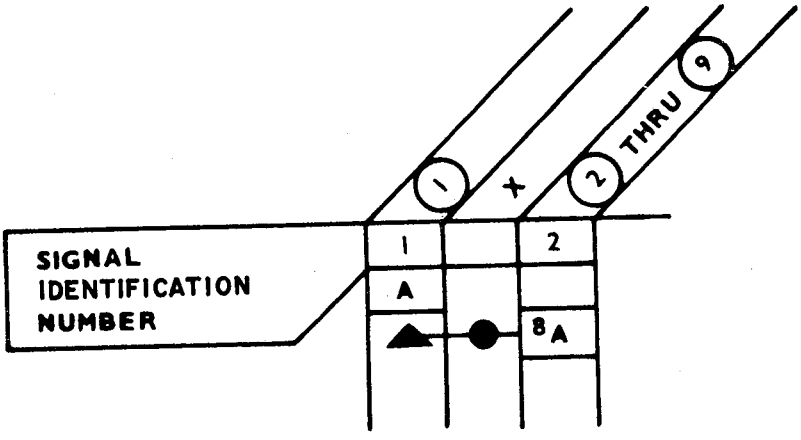
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:38Z
Check the source to verify that this is the current version before use.

The design outline (not contracted) would appear as follows:



SIGNAL IDENTIFICATIONS
1. WAVEFORM DATA
2. WAVEFORM DATA
3. WAVEFORM DATA
4. WAVEFORM DATA
5. WAVEFORM DATA
6. WAVEFORM DATA
7. WAVEFORM DATA
8. WAVEFORM DATA
9. WAVEFORM DATA

The design outline (contracted) would appear as follows:



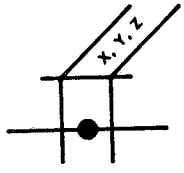
SIGNAL IDENTIFICATIONS

1. WAVE FORM DATA
2.

IDENT. NUMBER	EVENT MEASUREMENT POINT	WAVEFORM DATA
2.1	TP1	
2.2	TP2	
2.3	TP3	
2.4	TP4	
2.5	TP5	
2.6	TP6	
2.7	TP7	
2.8	TP8	

In this example, the eight ouput events of functional entity X are contracted into a single event. The small 8 in the upper left corner of the event box denotes the number of contracted events. Each of the contracted events is assigned a unique identification number as indicated in the signal identification area in the example above. The physical locations of the event measurement points are identified either in the heading or the event measure point column in the signal identifications area. Waveform data is provided for each of the events in the waveform data column in the signal identifications area.

30.7.4.3.6.2 Contracted functional entity. A contracted functional entity represents a number of single functional entities and is constructed as follows:

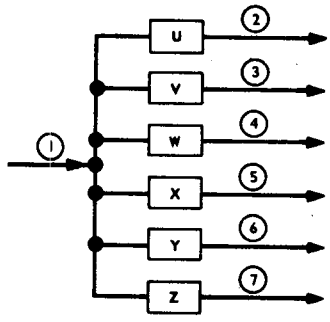


A contracted functional entity is identified by noting that more than one functional entity is listed in the heading corresponding to the column of which it is a part.

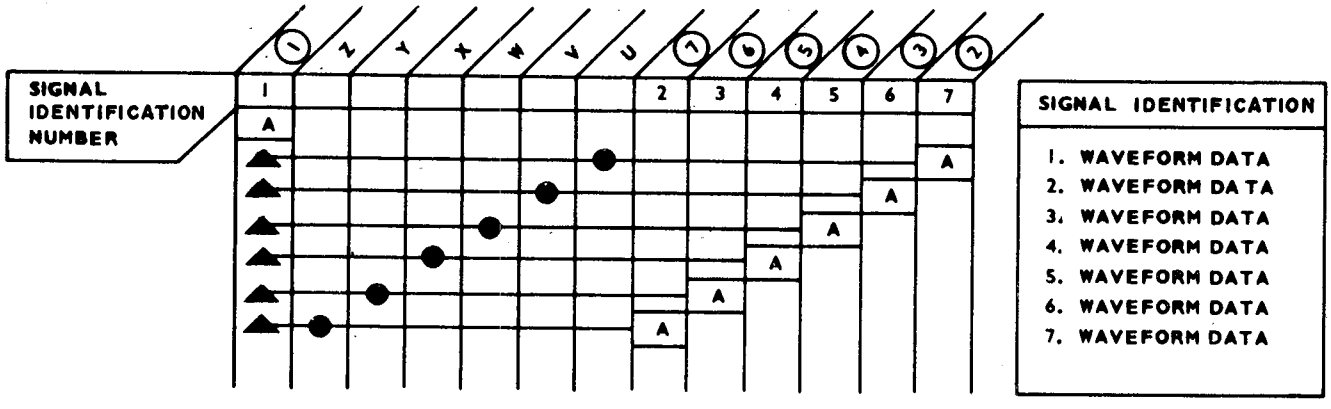
The following is an example of contracted functional entities. Consider the case in which a signal is distributed to six separate functional entities.

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The block diagram would appear as follows:

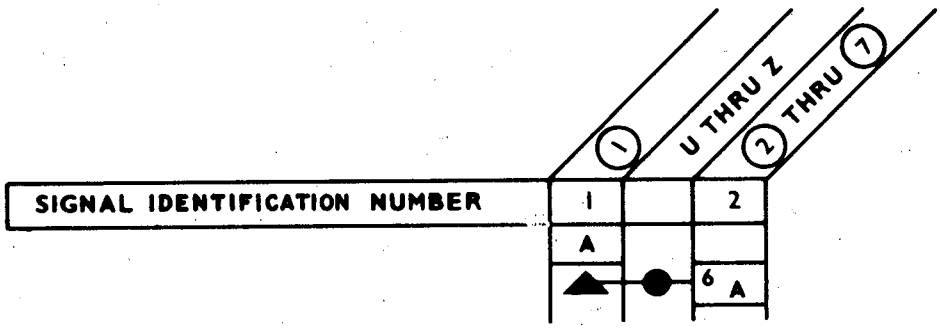


The design outline (not contracted) would appear as follows:



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The design outline (contracted) would appear as follows:



SIGNAL IDENTIFICATION

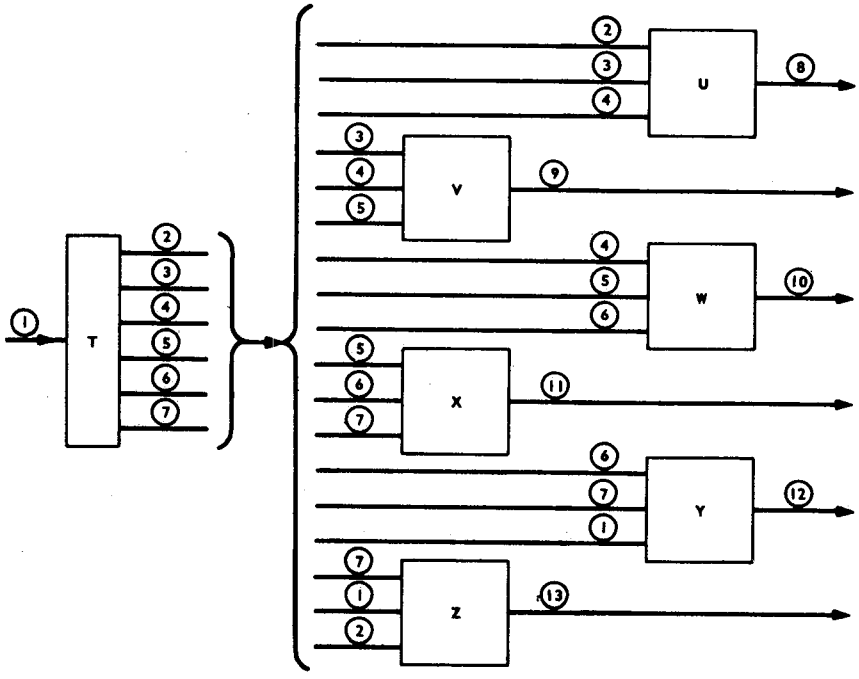
1. WAVEFORM DATA
2.

IDENT. NUMBER	EVENT MEASUREMENT POINT	WAVEFORM DATA	ASSOCIATED FUNCTIONAL ENTITY	λ	ERT
2.1	(2)		U		
2.2	(3)		V		
2.3	(4)		W		
2.4	(5)		X		
2.5	(6)		Y		
2.6	(7)		Z		

In this example, the six output events have been contracted into one event entry. In addition, the six functional entities associated with these events have been contracted into one entry. The identification number, measurement point, and waveform data of each contracted event are described in the signal identifications area in the same manner as in the event contractions example. The functional entity associated with each event is identified by an additional column in the signal identifications area as shown above. The failure rate (λ) and estimated repair time (ERT) associated with each functional entity is also provided.

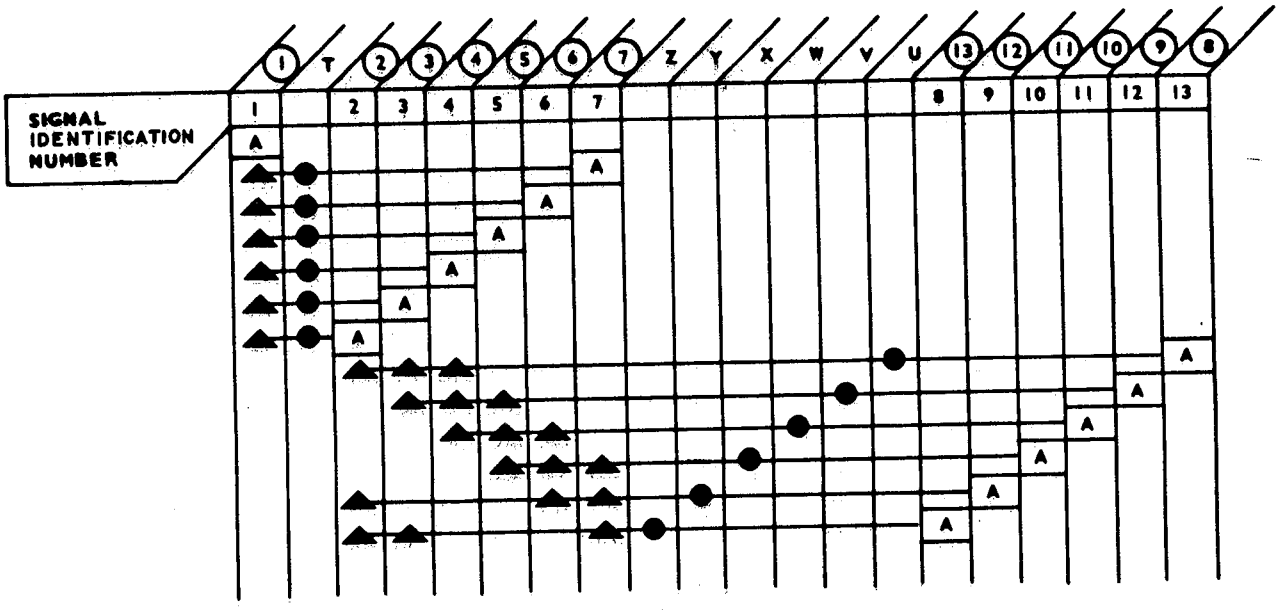
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30.7.4.3.6.3 Contracted dependency chains. The following example illustrates the mapping of a complex design resulting in a contracted dependency chain. Consider the case represented, by the following block diagram.



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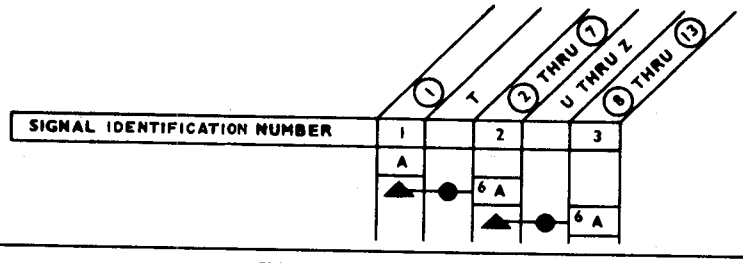
The design outline (not contracted) would appear as follows:



Signal Identifications
1. WAVEFORM DATA
2. WAVEFORM DATA
3. WAVEFORM DATA
4. WAVEFORM DATA
5. WAVEFORM DATA
6. WAVEFORM DATA
7. WAVEFORM DATA
8. WAVEFORM DATA
9. WAVEFORM DATA
10. WAVEFORM DATA
11. WAVEFORM DATA
12. WAVEFORM DATA
13. WAVEFORM DATA

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The following design outline (contracted) represents a contracted dependency chain.



SIGNAL IDENTIFICATION

1. WAVEFORM DATA

2.

IDENT. NUMBER	EVENT MEASUREMENT POINT	WAVEFORM DATA	IDENT. NUMBER	EVENT MEASUREMENT POINT	WAVEFORM DATA
2.1	②		2.4	⑤	
2.2	③		2.5	⑥	
2.3	④		2.6	⑦	

3.

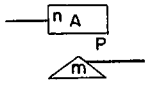
IDENT. NUMBER	EVENT MEASUREMENT POINT	WAVEFORM DATA	ASSOCIATED ENTITY	FUNCTIONAL λ	FUNCTIONAL ERT	REQUIRED INPUT EVENTS
3.1	⑧		U			2.1, 2.2, 2.3
3.2	⑨		V			2.2, 2.3, 2.4
3.3	⑩		W			2.3, 2.4, 2.5
3.4	⑪		X			2.4, 2.5, 2.6
3.5	⑫		Y			2.5, 2.6, 2.1
3.6	⑬		Z			2.6, 2.1, 2.2

In this example, the six output events and their associated functional entities have been contracted. In addition, the six outputs of functional entity T have been contracted. The identification number, measurement point, and waveform data associated with the event outputs of functional entity T are described in the same manner as in 30.7.4.3.6.3. The identification number, measurement point, waveform data, and functional entity associated with each of the output events at ⑤ through ⑬ are described in the same manner as in 30.7.4.3.6.2. In addition, the input events associated with each of the output events are tabularized as shown.

30.7.4.3.6.4 Event/dependency interface for contracted events. In the situation where several events are mapped from a contracted event, a dependency marker with special notation is used. Two types of

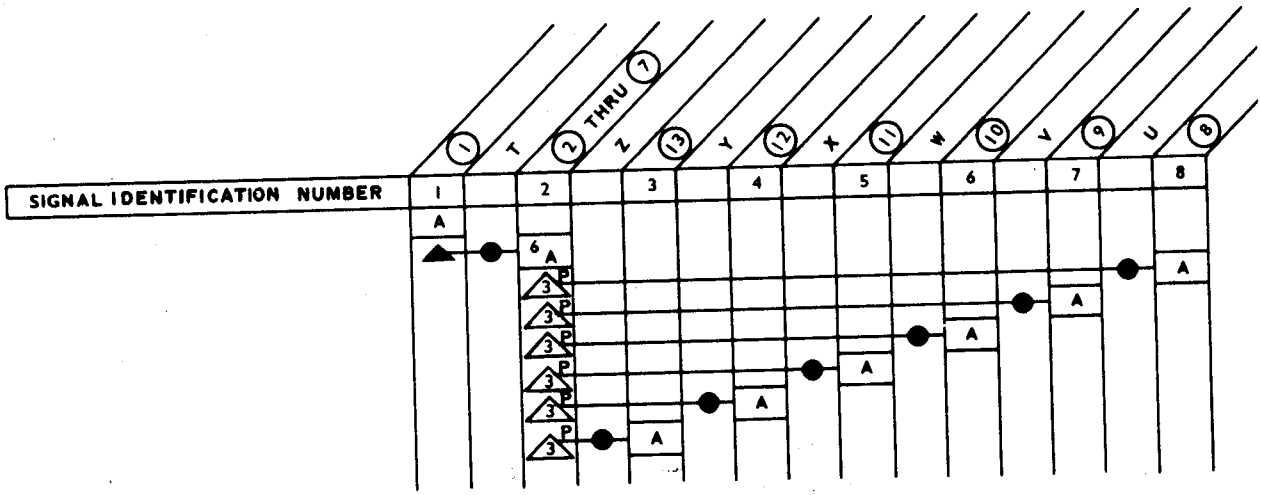
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:38Z
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special notations are used. The first type relates to particular combinations of events from a contracted event and is shown as follows:



The use of the m and p with the dependency marker indicates that m particular events out of the n contracted events are required to produce the ensuing event. The following example illustrates the use of this type of event/dependency interface.

Consider the block diagram of 30.7.4.3.6.3 in which only the inputs of functional entity T are contracted. Assume it is desirable from a disclosure viewpoint to not contract the outputs at ⑧ through ⑬ and their associated functional entities. The design outline would appear as follows:



SIGNAL IDENTIFICATIONS

1. WAVEFORM DATA

2.

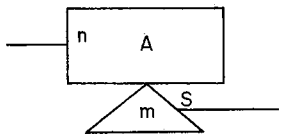
IDENT. NUMBER	EVENT MEASUREMENT POINT	WAVEFORM DATA
2.1	②	
2.2	③	
2.3	④	
2.4	⑤	
2.5	⑥	
2.6	⑦	

3. WAVEFORM DATA: REQUIRED INPUTS: 2.1, 2.2, 2.3
4. WAVEFORM DATA: REQUIRED INPUTS: 2.2, 2.3, 2.4
5. WAVEFORM DATA: REQUIRED INPUTS: 2.3, 2.4, 2.5
6. WAVEFORM DATA: REQUIRED INPUTS: 2.4, 2.5, 2.6
7. WAVEFORM DATA: REQUIRED INPUTS: 2.5, 2.6, 2.1
8. WAVEFORM DATA: REQUIRED INPUTS: 2.6, 2.1, 2.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:38Z
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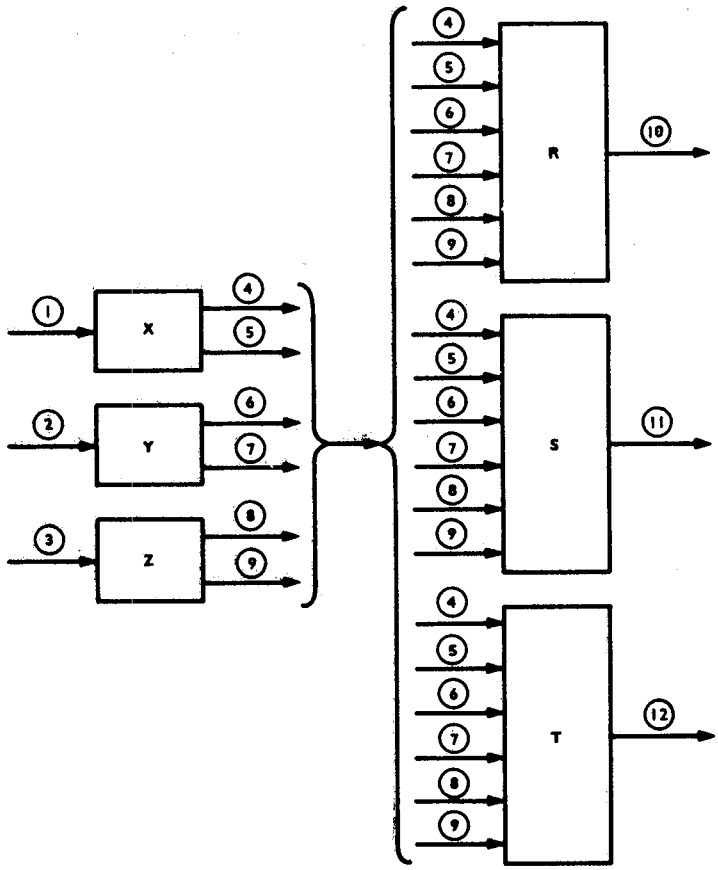
The event/dependency interface for the events at ⑧ through ⑬ indicates that three particular events of the contracted six events are required to produce the desired output events. The data for these events identifies the required three events as shown. See figures 18 and 19 for additional examples of this type event/dependency interface.

The second type of dependency marker notation relates to sets of combinations of events and is shown as follows:



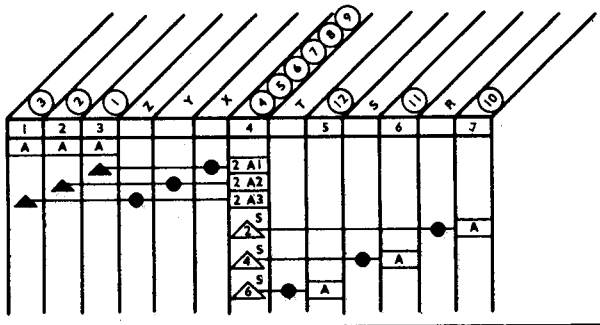
The use of the m and s' with the dependency marker indicates that m number of sets of events is required to produce the ensuing event. A given set is defined as the number of combinations of n events taken m at a time. ($N=C^n_m$). A different set of combinations is generated for each value of m(m<n).

The following example illustrates the use of the event/dependency interface for sets of combinations. Consider the case represented by the following block diagram.



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The following design outline represents the functional entities and the signal flow between X, Y, Z, and R, S, T.



SIGNAL IDENTIFICATIONS

1. WAVEFORM DATA
2. WAVEFORM DATA
3. WAVEFORM DATA

4

IDENT. NO.	EVENT MEAS. POINT	WAVEFORM DATA	ASSOC FUNC ENTITY
4.1	(4)		X
4.2	(5)		X
4.3	(6)		Y
4.4	(7)		Y
4.5	(8)		Z
4.6	(9)		Z

5

WAVEFORM DATA	REQUIRED INPUTS	$N = C_m^n = \frac{n!}{m!\,(n-m)!}$
	ALL SIX	1
OR	ANY FIVE	6
OR	ANY FOUR	15
OR	ANY THREE	20
OR	ANY TWO	15
OR	ANY ONE	6

6

WAVEFORM DATA	REQUIRED INPUTS	$N = C_m^n = \frac{n!}{m!\,(n-m)!}$
	ANY FIVE	6
OR	ANY FOUR	15
OR	ANY THREE	20
OR	ANY TWO	15

7

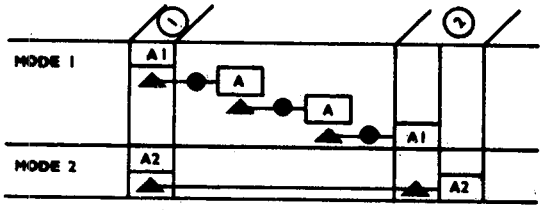
WAVEFORM DATA	REQUIRED INPUTS	$N = C_m^n = \frac{n!}{m!\,(n-m)!}$
	ANY TWO	15
OR	ANY FIVE	15

In this example the six signals from X, Y, and Z are described by the composite event designated by signal identification number 4. Signal number 5 on the signal identifications chart shows that the event at (12) is available when alternative sets of combinations occur. This means that in order for a signal to be available at measurement point (12) a combination of all six events must be available to T or any combination of five events must be available to T or any combination of four, etc. The column at the right indicates the total number of possible combinations for each set. Similarly, signal identification number 6 shows that the event at (11) is available when a combination of any five events are available to S and S is functioning properly, or any combination of four or three or two. The column at the right, as before, indicates the total number of possible combinations for any given set.

The dependency interface for sets of combinations becomes quite useful in the design of complex radar and sonar systems, as well as sophisticated digital equipments.

30.7.4.3.7 Event commonality. When a complex dependency chain produces an end availability which proves a group of functional entities good under one mode of operation, it may be proven good under another mode as follows:

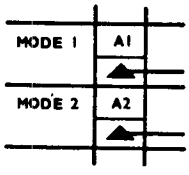
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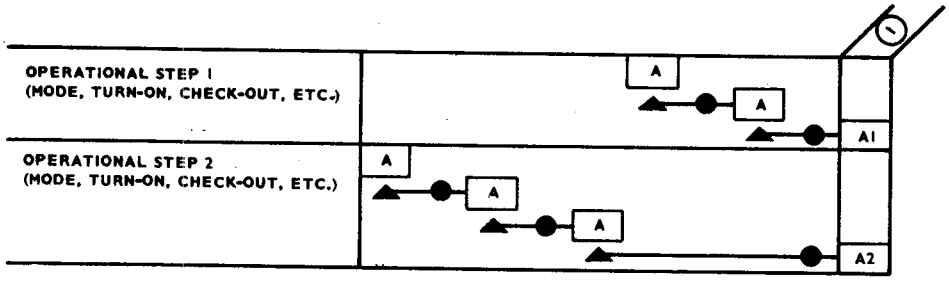
The two end availabilities above (furthest to the right) are measured at the same physical test point but chartered in two adjacent columns containing the same signal identification number. The last line is interpreted thusly: If A2 at ① is now available, and A1 at ② was available during mode 1, then A2 at ② is available for mode 2. This implies that all functional entities required for mode 1 are also necessary for mode 2. This technique eliminates the need for redeveloping the entire dependency chain of mode 1 again for mode 2.

A second case occurs when two events of different operational steps occur at the same measurement point but are a result of different dependency chains. The following two examples illustrate this situation.

Example 1:



Example 2:

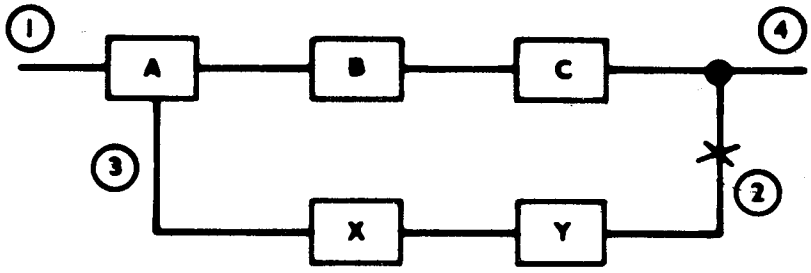


30.7.4.3.8 Feedback loops. When the output of a circuit is sent back to appear as part of the input to the circuit, a feedback loop exists. To disclose the functional dependency of such a loop requires that the loop be broken. Under these conditions, the normal (main) input into the loop, plus a proper stimulus inserted at the break, will result in a known output. Breaking the loop allows functional dependency to appear as a simple serial dependency chain. Feedback loops are identified in the procedure column and enclosed in a manner similar to check-out steps (refer to example feedback block diagram and design outline below). The complete dependency structure of the broken loop is then shown within the enclosed area. Directly below the enclosed area, the dependency structure of the loop is again shown but with the loop connected as normal. This structure appears as a horizontal line consisting of a dependency mark, a series of dots and an output.

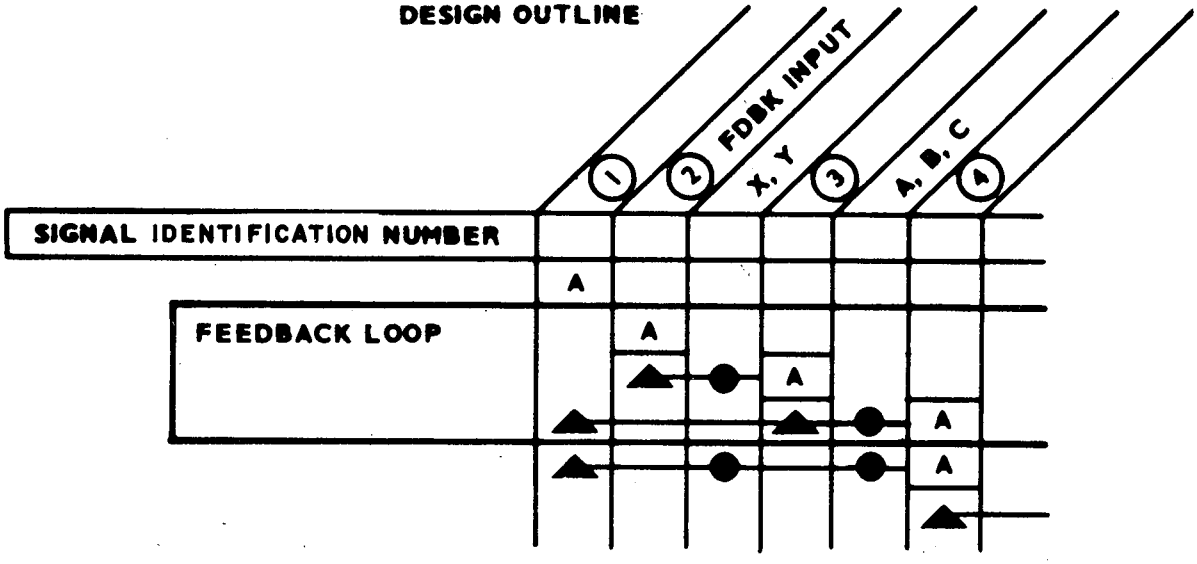
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T13:38Z
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In words, the line shows that if the input to the loop is available and all functional entities within the loop are good, then the output of the loop is available. Using this technique, the detailed dependency structure of a feedback loop can be disclosed while also showing how the loop as a whole fits into the general dependency scheme of the design outline when normally connected. Figure 12 also provides several examples of feedback loops.

FEEDBACK BLOCK DIAGRAM



DESIGN OUTLINE



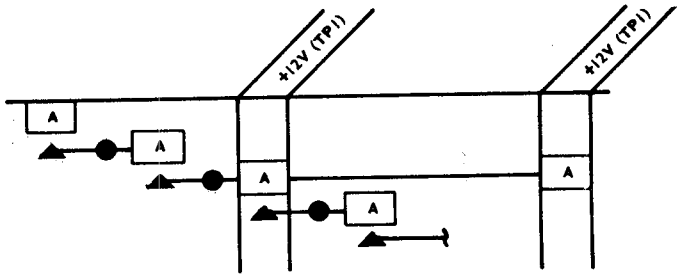
SIGNAL IDENTIFICATIONS
1. WAVEFORM DATA
2. FEEDBACK WAVEFORM DATA
3. WAVEFORM DATA
4. WAVEFORM DATA

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30.7.4.3.9 Re-entered event. Occasionally, design outline presentation is enhanced when an event is re-entered.

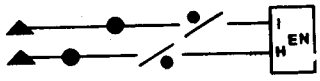
An event is re-entered by simply extending the horizontal line, of which it is a part, to the desired point of re-entry and again recording the event. All information associated with the event (heading, signal identification number, λ, ERT, etc.) must also be re-entered in the appropriate column.

An example of event re-entry is as follows:



In this example the +12V occurs in the middle of the design outline and is further involved in the dependency structure. Thus, it is entered where it is first developed in the dependency structure and then re-entered for further use.

30.7.4.3.10 Energized relay events. When relays are held energized by a path separate from that of the initial energizing path, a variation of a composite event box is used to represent the situation. The resulting dependency chain normally appears as follows:



The first line terminated by an "T" shows the path for initially energizing the relay. The second line terminated by an "H" identifies the holding path.

30.7.4.3.11 Time delays (see figure 13). Protective time delays which occur after an operational step is initiated are illustrated by placing a band of screen shading horizontally across the page immediately after the entry of the time delay device. Time is assumed to begin with the setting of the switch that initiates the step in which the time delay occurs. The length of time delay is indicated within the shaded band. When the time delay is not directly related to the step, special notations may be added within the time delay band - e.g. "2 seconds after motor exceeds 3000 rpm".

40. PRODUCTION TECHNIQUES

40.1 Shade option. The definition of physical and functional boundaries on the design disclosures can be greatly enhanced by the use of shading techniques. By using shades to define hardware and functional boundaries respectively, confusing boundary lines are eliminated, and subfunctionalization becomes more apparent.

At certain phases of the life cycle, the procuring agency may elect to print the design disclosure set and employ the shading techniques. The following paragraphs discuss a proven method for securing the shades.

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40.1.1 Gray shades. Shades of gray are used to define functional areas enclosed by a light solid line. The density of the shading is dependent on the degree of functionalization. The highest functional level (most general) is defined by the lightest gray shade. Each succeeding lower level (more detailed) of functionalization is shown by succeedingly denser shades of gray. This does not mean that a particular level of gray should be associated with a particular subfunctional level. The density of a shade only shows the degree of functionalization. Shades of gray will be used to define all functional areas including the blocked symbols on detailed block diagrams. However, logic symbols on logic block diagrams are to be considered in the same category as part symbols on blocked schematics. They will remain defined by solid lines but grouped by gray shades into functional entities. For example, on figure 9 the circuits within shift register (L-SR-1) will be shown but will be grouped by an overall gray shade enclosed by a solid light line. Shades of gray will also be used to define hardware boundaries enclosed by broken lines. The lightest gray shade will indicate the most general hardware containment with darker shades the more detailed. For example, on a detailed block diagram the lightest gray shade would represent the over-all equipment while the darker shades would represent lower levels of hardware containment.

40.2 Shade negative techniques.

40.2.1 Masks. Masks for the preparation of shaded areas may be prepared as follows: (Other techniques which produce the same results at equal or less cost are acceptable.) Basically, one screen tint mask is prepared for each level of shade. Therefore, a diagram having three levels of gray would have one mask cut for each individual shade level, making a total of three masks. These three masks plus the line negative constitute the complete package required by the printer to produce the black printing plates for the press run. One of the simplest techniques is as follows:

(a) Place the reduced size line negative on a light table.
(b) Place the register marks on the negative outside of the image area.
(c) Lay a piece of Rubylith D3R (Ulano Graphic Arts Supplies, Inc.) or equivalent over the negative and register it with the negative.
(d) Using an exacto knife or equivalent, strip away the area not to be shaded and leave opaque the shaded areas.
(e) Remove the mask and repeat the process for each level of shading required, starting with the first level of gray and ending with the darkest shade.
(f) Identify each mask by affixing a label which contains the art number of the line negative and the level of the shade - e.g. 1st level gray, 2nd level gray, etc.

40.2.2 Negatives. For each page a single lithographic negative should be prepared for the black image, a single lithographic negative for the gray shades. All negatives should be register-marked and identified such that no confusion can exist as to their relationship. Prior to printing or delivery, the negatives for the black image should be opaqued sufficiently to be free from all traces of cropmarks that were used to provide area boundaries for the making of the gray shade masks. Pinholes, likewise, should be removed.

40.2.2.1 Shade density. Shades of gray background indicate hardware grouping and hardware a dif- subordination. The gray shade background should be a black-screen tint of a different percentage density for each level using a 133 line, zero degree rotation screen.

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Level	Printed dot size Diameter 1/ (black)	Negative dot size Diameter 1/ (clear)
First	0.002-inch	0.002-inch
Second	0.004-inch	0.004-inch
Third	0.005-inch	0.005-inch
Fourth	0.006-inch	0.006-inch
Fifth	0.007-inch	0.007-inch

1/ ± 0.005-inch deviation allowable. Greater separation of tint is permissible when fewer than five levels are required on a diagram.

40.2.2.2 Technique of negative exposure to obtain shaded areas. The negatives the gray shades may be prepared by a contact process using a litho-ortho type film, a point source light, and a vacuum frame as follows: It should be noted that dimensions vary slightly from those on negatives and those on the printed page. The following stated dimensions are nominal.

(a) Place the unexposed film and a half-tone screen in a vacuum frame with the screen between the glass and the film.
(b) Register and key all masks together on the outside of the glass of the vacuum frame to ensure proper registration between masks and film.
(c) With all masks removed, expose the film to a point source of light. The exposure time for the gray shade negative should be sufficient to produce a 0.007-inch diameter clear dot. This step establishes the fifth level shades.
(d) Place the fifth-level mask on the registration key pins outside the vacuum frame.
(e) Expose gray-shade negative to the point source sufficiently to increase the dots in the unmasked area of the screen negative to a dot diameter such that a clear dot diameter of 0.006 inches is obtained. Steps (d) and (e) establish fourth-level shades.
(f) Remove the fifth-level mask and position the fourth-level mask.
(g) Expose gray-shade negatives sufficiently to increase the dots in the unmasked area of the shade negative to a diameter such that a clear dot diameter of 0.005 inches is obtained. Steps (f) and (g) establish third level shades.
(h) Subsequent exposures for the gray shade negatives are equivalent to the previous steps except for substitution of appropriate masks and the additional exposures to produce dot diameters (see 40.2.2.1) for dot diameter required by additional shade levels.

The following considerations are offered and may assist in obtaining better controls and uniformity in shade negatives:

(a) Pre-wet the film in water prior to immersion in the developer bath. (This should help preclude uneven development and developer streaks).
(b) A somewhat diluted developer and a slightly reduced developer temperature (66° in lieu of 68°F) will lengthen development time and the exactness of the precise time of removal of the film from the developer will be minimized.
(c) Place the pre-wet film in the developer, emulsion side up, making sure that it is totally submerged immediately and that no air bubbles adhere to the emulsion side. On the surface of the developer (tray development) there frequently exists patches of developer concentrate. Inserting the film emulsion side up will break up these patches and possible localized effects will be minimized.

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40.2.2.3 Screen negative quality. Screen density should be as specified in 40.2.2.1. An Eastman density comparator wedge, or equivalent, should be used to assure that specified dot diameters are maintained between shades. The use of a 21 step transparent gray scale wedge placed on an edge of the film (in the vacuum frame) but outside of the screen or image area will to some degree offset the effect of so many of the variables involved in producing screen negatives. Developing the film to the point where the desired wedge of the gray scale first appears should give reasonably uniform results.

40.2.3 Printing. The black line negative and the gray shade negative should be burned onto the same plate before printing to eliminate an additional press run.

40.2.4 Early phase review deliveries. For early phase deliveries the procuring agency may elect, for economic reasons, to allow delivery of upsize (original) artwork. For this purpose, pencil layouts may be used along with hand lettering which is legible. Companion text diagrams must still be typed, but do not have to conform to diagram layout.

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